

External Validation of Artificial Intelligence for Detection of Heart Failure with Preserved Ejection Fraction

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Akerman et al assessed the diagnostic and prognostic performance of a AI HFpEF model based on TTE imaging, and compared it against H2FPEF and HFA-PEFF. The assessed model (EchoGo) has been developed and validated in a previous study. In the current study, the authors provided additional validation in an independent cohort of 240 HFpEF cases and 256 matched controls with clinically indicated TTEs. The AI model had higher sensitivity and specificity than the scores. I commend the authors for the study, as improved tools to diagnose HFpEF are greatly needed.

I have the following comments for the authors to consider:

1. In my view, the introduction might be improved to better outline the rationale and need for the study. The assessed AI HFpEF model has already undergone initial validation, and been assessed for both diagnostic and prognostic performance. I suggest that the introduction should provide the rationale for the study in that context.
2. The introduction states "Heart failure (HF) with preserved ejection fraction (HFpEF), impacting up to 5.5% of the general population [...]", which stands out as a high prevalence estimate. This estimate appears to have originated from a 1999 survey of patients aged >70 years, and is hardly representative of the general population.
3. Approximately 20% excluded due to insufficient image quality. If one were to take a sceptical position, one might count these towards being "non-diagnostic" by the AI HFpEF model in a practical sense, in which case one of the discussed advantages vs. the clinical models would be diminished. This might be discussed as a limitation.
4. As I understand, missingness for variables needed for the clinical scores (e.g. NT-proBNP) would lead to a classification as "non-diagnostic". It seems that this approach might overestimate "non-diagnostic" rates of the clinical scores as they would be used in a clinical setting, where one might in those cases simply acquire the needed data.
5. Were any steps taken to rule out patients with previous HFrEF who had now improved to the preserved EF range?

Reviewer #2

(Remarks to the Author)

In this study, Ackerman et al. perform an external validation of a deep learning model to identify HFpEF using transthoracic echocardiogram (TTE) images (AI HFpEF) utilizing a retrospective case/control design in roughly 500 total patients. The AI HFpEF score discriminates HFpEF (AUROC 0.798), but performance is substantially lower than the original validation study (0.95), and performance here is comparable to clinical score comparators. There is some evidence presented suggesting that AI HFpEF may complement clinical scores. AI HFpEF predictions are associated with longitudinal mortality and/or HF hospitalization.

True external validation of AI models is very important and I applaud the authors for taking it on. However, the impact of the current study is likely to be limited by small sample size, substantial reduction in performance compared to the original validation study, and some issues in the analysis that raise questions and I believe tend to overemphasize the value of the AI HFPEF model in the current setting. Specific comments below:

Major

1. The performance of AI HFpEF here (AUROC 0.798) is substantially lower than that observed in the original derivation/validation study (0.95). Substantially more discussion is warranted regarding potential causes and implications of this generalization cost.
2. Much emphasis is placed on the lower rate of 'non-diagnostic' results using AI HFpEF versus the clinical scores, but I think these comparisons are a bit artificial at least for H2FPEF. I understand that the H2FPEF publication promoted a score ≥ 6 as high probability for simplicity and as a practical value for targeting high PPV, but since the score translates into a continuous probability it is not as though scores classified here as 'indeterminate' (2-5) provide no information. Indeed, we know from the AUROC and curves that both AI HFpEF and H2FPEF actually contain the same amount of predictive information overall. I believe a fairer comparison would be to compare AI HFpEF with H2FPEF when both are thresholded at the same probability value (e.g., 0.5).
3. More information is needed regarding inclusion criteria and outcomes
 - a. How was the 2018-2022 window chosen?
 - b. Why limit to 300 cases to match (rather than all the potential cases)? How was 300 determined?
 - c. More detail is needed regarding "technically evaluable images". What does this mean? Note that 20% of individuals in each group were excluded for this vague criterion. It is also curious why this criterion was applied downstream of the case/control matching...why wouldn't the image quality criterion be applied early on so that 300 cases and controls could be robustly secured?
 - d. Almost half of the TTEs were excluded for missing biplane LVEF. This seems to be a large proportion. Is biplane LVEF measurement standard at this institution? If not, why was it chosen as a selection criterion? Are there temporal/trends in practice (or other potential indirect mechanisms of selection) that need to be considered here?
 - e. Controls had "no evidence of HF within 1-year prior to and after the index TTE". Does this mean no inpatient HF code in this timeframe? Or any HF code? Or any evidence of HF as determined by the manual chart review?
 - f. "Presence or absence of HF hospitalization was additionally adjudicated at the time of manual chart review". Statement is vague. Were all cases and controls manually reviewed? What criteria were applied to define HF status using manual chart review (both for the purposes of defining HF case status and the HF outcome)?
 - g. Were the adjudicators blinded to HFpEF case/control status when ascertaining clinical variables and outcomes?
4. The confidence intervals for AUROC (e.g., AI HFpEF 0.798, 95% CI 0.756-0.799) seem off. I understand AUROC CIs can be asymmetric, but the upper bound being 0.799 when the effect estimate is 0.798 and the sample size is modest seems questionable. Likewise for H2FpEF (AUROC 0.788, 95% 0.745-0.798).
5. BNP/NTproBNP appears to be missing for >25% of HFpEF cases. This seems unexpected for patients with HF as a primary diagnosis for hospitalization. Could the authors comment?
6. In the discussion, the authors envision a clinical use case in which AI HFpEF is applied for individuals with indeterminate clinical risk scores. However, I believe the converse (applying clinical risk scores if/only if AI HFpEF is indeterminate) is more practical, as it minimizes the number of individuals who would require complex clinical score availability/calculation, which the authors argue is an important goal.
7. Limitations, "although biplane LVEF values were not verified...AI HFpEF computed biplane LVEFs". This is the first mention of this process that I can see. How were these computed values utilized in the selection process?
8. I believe the Discussion would be improved by better focus on the specific use case of AI HFpEF and where the specific added value here. Several points regarding the challenges of HFpEF (absence of a standard definition, heterogeneity of etiology) that are discussed are not addressed by AI HFpEF.
9. I don't find the Alluvial plots in Figure 3 to be particularly useful. They summarize reclassification but (unless I am misinterpreting them) they do not indicate whether reclassification is appropriate or inappropriate. I believe a categorical net reclassification analysis would be more customary (where non-diagnostic could be considered an intermediate class), and the corresponding tables would give a better sense of whether the reclassification is correct or not. Along the same lines I think presentation of continuous NRI or integrated discrimination improvement (IDI) using the continuous probabilities (i.e., AI HFpEF and H2FPEF) would be useful information.
10. Some comments regarding the decision curve analysis
 - a. More information is needed in the caption as many readers may not be familiar with this type of analysis. The information in the Supplemental Methods is useful but much of it should be incorporated in the caption or elsewhere in the main Methods.
 - b. As currently parameterized, I believe the "Either" and "Combined" approaches are identical. This is supported by the fact the curves are entirely overlapping. That said, as per my comment above I would consider adding a condition where one runs the clinical score only if AI HFpEF is indeterminate.
 - c. I would recommend the x-axis is expanded to the point where the risk stratification approaches intersect with "Treat None" – as this depicts the threshold where risk stratification is no longer useful (a useful data point to show).
 - d. Since the authors propose a risk threshold based on the DELIVER trial, I think it would be valuable to report the actual net benefit values and net interventions avoided for each strategy at that threshold.
11. Figure 5 (KM curves) – Why do the curves not start at 1?
12. The AI HFpEF model appears to display an unexpected behavior such that that true risk appears to decrease from quartile 2 to quartile 3. This is visible in the calibration plots (where quartile 3 is particularly miscalibrated) as well as Supplemental Table 7 where the HRs for quartile 3 versus quartile 1 are consistently lower than those for quartile 2 versus quartile 1. This seems to suggest an issue with model predictions in this range and requires further discussion/explanation.

Minor

1. In general there is too much discussion regarding the nuances of WPW, which is a rare condition, particularly when compared to the other abnormalities tested (RBBB, LBBB, etc).
2. Results, line 236 – not clear what the unit of these effect sizes are. Is it per-quartile increase?

3. Discussion, line 314 and elsewhere in the paragraph – patients were not really ‘enrolled’ in this retrospective study. Would rephrase.
4. Conclusions, line 330 – “higher accuracy than clinical scoring systems to classify patients...due to non-diagnostic rates from clinical scores”. I do not think the current results support such a statement from an “accuracy” standpoint. Would rephrase.
5. Discussion points regarding complementarity of AI and clinical information would be improved by referencing related work suggesting similar conclusions in areas such AF prediction (e.g., PMID 34743566) and CAD risk estimation (e.g., PMID 37700032) using ECG.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

I thank the authors for responding to my previous comments. I have only few comments remaining:

- NPV and PPV are dependent on the prevalence in the study population, which here is up-sampled due to the case-control design – as far as I understand these numbers were not adjusted for expected real-world prevalence, which hinders direct translation of NPV/PPV to the clinical setting and should be clarified for the reader.

- Abstract: “Models with continuous probability outputs (AI and H2FPEF) demonstrated similar discrimination and calibration, but the AI HFpEF model had higher sensitivity (AI: 77.4% vs. H2FPEF: 53.9% vs. HFA-PEFF: 63.2%) and specificity (50.2% vs. 12.8% vs. 29.0%), largely attributable to higher non-diagnostic rates (15.1% vs. 61.7% vs. 54.2%), even after accounting for missing multivariable inputs.” Should this not be “lower” non-diagnostic rates, as it refers to the AI HFpEF model?

- Results – Patient Population (page 9, row 190): “A final sample of 240 cases and 256 were identified after application of exclusions (Figure 1).” The sentence appears to be missing the word “controls”.

Reviewer #2

(Remarks to the Author)

The manuscript is improved by the changes made in revision, and I feel the value of the contribution is more apparent. I still have several concerns which I consider substantive:

1. I still feel that findings secondary to categorization/dichotomization are suboptimal or even misleading. Part of my motivation for raising these issues is that I feel the key study findings remain valuable without much of what I perceive to be an over-emphasis on arbitrary categories of continuous scores.
 - a. The H2FPEF score is a continuous score with a nearly identical AUROC to AI HFpEF. Therefore, claims of “higher classification performance” are not sufficiently justified. Higher sensitivity/specificity using AI-HFpEF versus H2FPEF is based purely on the operating points evaluated, and would differ across other operating points on the ROC curve (and on average, we know the ROC curves are the same, so there are also clearly points where sens/spec are higher for H2FPEF). Although I don’t think it makes a very strong argument, I think the authors can elect to keep the current evaluation at the cutpoints provided because the cutoffs are indeed mentioned in the H2FPEF paper, but the comparisons need to be placed in a more explicit and nuanced context that the comparison model could easily be compared at other cutpoints (and indeed, when they are compared using deciles of probability instead, performance is nearly identical, as expected). It would perhaps be another story if there were firm medical consensus or guideline statements promoting a H2FPEF score of 6-9 to make treatment decisions, but I just don’t see that as the case. For HFA-PEFF, which is not a continuous score, the comparison made is less problematic, but the analysis would still benefit from more context on the cutpoints themselves.
 - b. On a related note, I find the criteria used to define the categories of AI-HFpEF not sufficiently justified. How were these very particular (e.g., uncertainly less than 0.44, or prediction class probability greater than or equal to 0.5) cutpoints derived? This may be explained in the prior papers, but some more context in this paper/supplement is needed.
2. How are the test characteristics (e.g., sensitivity, specificity) calculated with respect to the ‘non-diagnostic’ categories? Is ‘non-diagnostic’ considered a negative test result? A robust way to handle this would be to calculate test characteristics considering all ‘non-diagnostic’ results are negative, and conversely all of them as positive.
3. I struggle in general with the term ‘non-diagnostic’ here. As the authors mention, there is no real gold standard for HFpEF so none of the scores or their values are ‘diagnostic’ in the first place. Furthermore, the intermediate scores still provide some information. Looking back at the original H2FPEF and HFA-PEFF papers, the term ‘intermediate’ appears to be employed and I think this is a far more appropriate term. Along the same lines, I think there needs to be a clearer distinction between ‘score not calculable’ and ‘intermediate value’, where the former is more truly in a sense ‘non-diagnostic’.
4. On a related note, in the discussion the authors seem to equate intermediate score values with ‘conflicting’ clinical information, but that is not necessarily the same. Someone may have an intermediate H2FPEF score simply because they are young...I don’t think that is necessarily in ‘conflict’ with the other risk factors.
5. More information is needed regarding the NRI analyses. I am very surprised to see highly positive NRI for AI-HFpEF versus H2PEF (NRI 0.40 [0.23-0.57] and IDI 0.07 [0.01-0.13]) when the AUROCs are the same. Further, the NRI shown in Fig 3 (0.03 [-0.06-0.20]) is very difficult and more expected. I’m unclear on what is different. Also, it is necessary to provide both the NRI+ and NRI- components given they imply very different things.

6. Since both scores are mis-calibrated, the overall NRI can be misleading. A percentile-based NRI would be a valuable addition (<https://pubmed.ncbi.nlm.nih.gov/29304237/>) to assess whether the NRI may be driven by asymmetric mis-calibration.

7. It would be an interesting and potentially fairly valuable secondary analysis to assess whether AI-HFpEF maintains a degree of classification performance among individuals with low clinical score values. This would suggest that the algorithm is picking up things that are not obvious and speaks more directly to added value.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

I thank the authors for addressing my previous remarks. I have no further comments.

Reviewer #2

(Remarks to the Author)

I am satisfied with the revisions made. Thanks to the authors and congratulations.

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Response to Reviewer 1:

We would like to thank the reviewer for their thoughtful comments which we feel have greatly strengthened our manuscript.

Akerman et al assessed the diagnostic and prognostic performance of a AI HFpEF model based on TTE imaging, and compared it against H2FPEF and HFA-PEFF. The assessed model (EchoGo) has been developed and validated in a previous study. In the current study, the authors provided additional validation in an independent cohort of 240 HFpEF cases and 256 matched controls with clinically indicated TTEs. The AI model had higher sensitivity and specificity than the scores. I commend the authors for the study, as improved tools to diagnose HFpEF are greatly needed.

We would like to thank the reviewer for their kind words.

1. In my view, the introduction might be improved to better outline the rationale and need for the study. The assessed AI HFpEF model has already undergone initial validation, and been assessed for both diagnostic and prognostic performance. I suggest that the introduction should provide the rationale for the study in that context.

Thank you for this important comment. We apologize that we did not fully describe the rationale for the current validation study in the context of the original manuscript and initial development and validation (Akerman, *JACC Adv*, 2023). Several items motivated the need for the current external validation. Firstly, between the original publication of the AI HFpEF model and now, there have been substantial updates to the model (e.g. V2 vs. V1) thus necessitating additional validation to ensure that model V2 performance is similar to V1. Secondly, the prior internal validation was multi-center but conducted across sites of a single integrated healthcare delivery network, the Mayo Clinic. As such, practice patterns for diastology grading may be more similar between sites affiliated with Mayo than non-Mayo sites, necessitating the need to validate performance in a truly external cohort. Finally, overall, we feel that there is scientific value to further external validation regardless of prior validation efforts in order to more fully assess the performance of this model in practice. This is in keeping with FDA guidance on the updating and incremental validation of models in clinical use (<https://www.fda.gov/files/medical%20devices/published/US-FDA-Artificial-Intelligence-and-Machine-Learning-Discussion-Paper.pdf>). We have modified the manuscript accordingly to reflect these comments.

Page 4, line 2 now reads:

*While this AI HFpEF model has previously undergone extensive multicenter internal validation across a large integrated delivery network,¹⁷ the motivation for the current study was to examine whether performance generalizes between institutions, and evaluate how architectural improvements due to experiences in clinical implementation (**Supplemental Methods**), may have impacted model performance.*

Supplemental Methods now reads:

Changes from Version 1 to Version 2 of the EchoGo Heart Failure AI Model

*The second version of the AI model iterated on the basic architecture described previously (Akerman, *JACC Adv*, 2023), and described in detail in Cassianni et al., *J Am Soc Echocardiogr*, 2024. The intention of the changes to the model were to provide more detailed, yet intuitive outputs to users. Specifically, upon clinical implementation, users requested a continuous class probability to be reported to help in the decision-making process, and more intuitive understanding of the non-diagnostic classification (i.e., uncertainty in the model classification and instability in the acquired image pertaining to the prediction). Lastly, version 2 implemented minor differences to the hyperparameter tuning, and expanded pre-processing and*

augmentation. Specifically, added pre-processing in version 2 standardized the input image (size, color, normalization of intensity, and framerate), and augmentations of the image (horizontal and vertical translations, rotations, zooming, and flipping); these additions were chosen due to experience during clinical implementation, with a goal of providing more a generalizable and robust model.

In version 1 of the AI model, outputs were categorical; input videos were labelled as either suggestive or not suggestive of heart failure with preserved ejection fraction (HFpEF), or inconclusive. Inputs were associated with a prediction class probability and expected entropy (uncertainty), with the categorical outputs associated with a decision threshold for prediction class probability (0.5) and expected entropy (0.59);

- 1) **Suggestive:** Prediction class probability greater than or equal to 0.5 and expected entropy below 0.59
- 2) **Not suggestive:** Prediction class probability greater less than 0.5 and expected entropy below 0.59
- 3) **Non-diagnostic/Inconclusive:** Expected entropy greater than or equal to 0.59, regardless of prediction class probability.

In version 2 of the AI model, the categorical output was retained, but a prediction class probability (“EchoGo Score”) was also reported alongside. A calibration function was added to version 2 of the AI model to ensure an accurate probability score. The inconclusive categorical report was also altered such that it represented model uncertainty and instability rather than the expected entropy (version 1). Model instability represented the standard deviation of all predictions for a given input videoclip, and model uncertainty represented 1 minus the maximum prediction class probability. Consistent with version 1, there were decision thresholds incorporated into the categorical reporting which related to prediction class probability (0.5), instability (0.20), and uncertainty (0.44);

- 1) **Suggestive:** Prediction class probability greater than or equal to 0.5, and instability less than 0.20 and uncertainty less than 0.44
- 2) **Not suggestive:** Prediction class probability less than 0.5, and instability less than 0.20 and uncertainty less than 0.44
- 3) **Non-diagnostic/Inconclusive:** Instability greater than or equal to 0.20 or uncertainty greater than or equal to 0.44, regardless of prediction class probability.

2. The introduction states “Heart failure (HF) with preserved ejection fraction (HFpEF), impacting up to 5.5% of the general population [...]”, which stands out as a high prevalence estimate. This estimate appears to have originated from a 1999 survey of patients aged >70 years, and is hardly representative of the general population.

Thank you for this point. We agree that this is an outdated statistic. We have now included data from the updated AHA/ACC 2024 Heart and Stroke Statistics and 2024 HF Statistics for Epidemiology and Outcomes.

Page 3, line 2 now reads:

Heart failure (HF) is a common and morbid disease impacting an estimated 56.2 million individuals worldwide with rising prevalence,¹ and impacting approximately 1 in 4 adults in their lifetime.² Of those suffering from HF, approximately half have a preserved left ventricular ejection fraction (HFpEF).³⁻⁶

3. Approximately 20% excluded due to insufficient image quality. If one were to take a sceptical position, one might count these towards being “non-diagnostic” by the AI HFpEF model in a practical sense, in which case one of the discussed advantages vs. the clinical models would be diminished. This might be discussed as a limitation.

Thank you for this comment. We agree that this represents an important limitation. As it is expected that approximately 10-20% of individuals have technically difficult echocardiographic windows (Fraiche, *J Am Soc Echocardiogr*, 2020), this proportion is within the range of expected values. However, this is also an important caveat to the discussion surrounding non-diagnostic outputs, and the impact it might have on performance comparisons. We have now acknowledged this as a limitation as follows.

Page 17, line 10 now reads:

Second, categorical outputs from all three models incorporate a non-diagnostic classification intended to support further confirmatory testing; direct comparison between models therefore must consider whether non-diagnostic outputs are due to missingness or discordance, and whether this would occur in clinical implementation. For instance, NT-proBNP was missing in nearly a quarter of patients, reflecting the challenges of guideline adherence in clinical practice as well as the clinical scores that utilize such laboratory markers, but could be obtained as required. When non-diagnostic outputs were considered a relevant element in the decision-making process, the current study demonstrated superiority of the AI HFpEF model. However, it must also be acknowledged that several patients were excluded due to poor image quality. While this degree of poor image quality is within published norms for echocardiography,⁴¹ it nevertheless exerts a greater effect on the AI HFpEF model compared to multiparametric clinical scores.

4. As I understand, missingness for variables needed for the clinical scores (e.g. NT-proBNP) would lead to a classification as “non-diagnostic”. It seems that this approach might overestimate “non-diagnostic” rates of the clinical scores as they would be used in a clinical setting, where one might in those cases simply acquire the needed data.

Thank you for this comment. While we agree that the missingness of NT-proBNP in this study may overestimate the rate of non-diagnostic results compared to a setting where NT-proBNP is measured in all individuals, we feel that the low rate of NT-proBNP testing may nonetheless be reflective of actual practice where testing is not universal. While the missingness of NT-proBNP would not immediately result in a non-diagnostic outcome (i.e., it is only one part of the calculation for the HFA-PEFF score), it is important to note that it will have a large impact on the outputs for this particular model. We also understand that it is important to acknowledge how this might or might not reflect clinical practice where a clinician might have access to historical data, but instead, we are interested in how the information put forward to a clinician might impact their decision-making process, including the choice to run further tests.

We have added this as a limitation and discussed the contextualization further in the following section.

Page 17, line 10 now reads:

Second, categorical outputs from all three models incorporate a non-diagnostic classification intended to support further confirmatory testing; direct comparison between models therefore must consider whether non-diagnostic outputs are due to missingness or discordance, and whether this would occur in clinical implementation. For instance, NT-proBNP was missing in nearly a quarter of patients, reflecting the challenges of guideline adherence in clinical practice as well as the clinical scores that utilize such laboratory markers, but could be obtained as required. When non-diagnostic outputs were considered a relevant element in the decision-making process, the current study demonstrated superiority of the AI HFpEF model. However, it must also be acknowledged that several patients were excluded due to poor image quality. While this degree of poor image quality is within published norms for echocardiography,⁴¹ it nevertheless exerts a greater effect on the AI HFpEF model compared to multiparametric clinical scores.

5. Were any steps taken to rule out patients with previous HFrEF who had now improved to the preserved EF range?

Thank you for this important point. For the majority of individuals in the cohort, their index echocardiogram was the first at our medical center and thus it is not possible to exclude that they had prior HFrEF with improvement. As such, the results should be seen as generalizable to the population with a recovered LVEF. We have added this to the limitations as follows.

Page 17, line 2 now reads:

First, it must be acknowledged that the clinical HFpEF syndrome has no clear and consistent definition, and a heterogenous etiology, meaning that not all HFpEF phenotypes and definitions will be captured in the current study. While the definitions adopted herein are consistent with recent HF guidelines,⁴⁰ other patients who might be reasonably associated with the HFpEF syndrome might not be adopted under the current definition; (i) patients in whom current (or prior) ejection fraction values are below 50%, (ii) left ventricular filling pressure confirmed via alternative methods such as exercise echocardiography or right heart catheterization, and (iii) patients who might have been hospitalized due to HF at different clinical sites, or captured at different phases of the diagnostic pathway (e.g., POCUS or advanced HF clinics).

Response to Reviewer 2:

We would like to thank the reviewer for their thoughtful comments which we feel have greatly strengthened our manuscript.

1. The performance of AI HFpEF here (AUROC 0.798) is substantially lower than that observed in the original derivation/validation study (0.95). Substantially more discussion is warranted regarding potential causes and implications of this generalization cost.

Thank you for this important comment. There are several reasons for the drop in performance observed. First, between the original publication of the AI HFpEF model (Akerman et al. *JACC Adv*, 2023) and now, there have been substantial updates to the model (e.g. V2 vs. V1) that could have impacted performance. Second, the prior internal validation was multicenter but conducted across sites of a single integrated healthcare delivery network, the Mayo Clinic. As such, practice patterns for diastology grading may be more similar between sites affiliated with Mayo than non-Mayo sites, resulting in a drop in performance when applied to a true external cohort. Third, the drop in AUROC from internal to external validation of this AI HFpEF model is similar to that observed in other models, where performance in the external validation cohort is inferior due to model optimism and overfitting during derivation.

We agree with the reviewer that the implications of this generalization cost are important, hence one of the motivations for the current external validation effort. We have modified the introduction and discussion accordingly to address this performance drop.

Page 4, line 2 now reads:

*While this AI HFpEF model has previously undergone extensive multicenter internal validation across a large integrated delivery network,¹⁷ the motivation for the current study was to examine whether performance generalizes between institutions, and evaluate how architectural improvements due to experiences in clinical implementation (**Supplemental Methods**), may have impacted model performance.*

Supplemental Methods now reads:

Changes from Version 1 to Version 2 of the EchoGo Heart Failure AI Model

*The second version of the AI model iterated on the basic architecture described previously (Akerman, *JACC Adv*, 2023), and described in detail in Cassianni et al., *J Am Soc Echocardiogr*, 2024. The intention of the*

changes to the model were to provide more detailed, yet intuitive outputs to users. Specifically, upon clinical implementation, users requested a continuous class probability to be reported to help in the decision-making process, and more intuitive understanding of the non-diagnostic classification (i.e., uncertainty in the model classification and instability in the acquired image pertaining to the prediction). Lastly, version 2 implemented minor differences to the hyperparameter tuning, and expanded pre-processing and augmentation. Specifically, added pre-processing in version 2 standardized the input image (size, color, normalization of intensity, and framerate), and augmentations of the image (horizontal and vertical translations, rotations, zooming, and flipping); these additions were chosen due to experience during clinical implementation, with a goal of providing more a generalizable and robust model.

In version 1 of the AI model, outputs were categorical; input videos were labelled as either suggestive or not suggestive of heart failure with preserved ejection fraction (HFpEF), or inconclusive. Inputs were associated with a prediction class probability and expected entropy (uncertainty), with the categorical outputs associated with a decision threshold for prediction class probability (0.5) and expected entropy (0.59);

- 4) **Suggestive:** Prediction class probability greater than or equal to 0.5 and expected entropy below 0.59
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In version 2 of the AI model, the categorical output was retained, but a prediction class probability (“EchoGo Score”) was also reported alongside. A calibration function was added to version 2 of the AI model to ensure an accurate probability score. The inconclusive categorical report was also altered such that it represented model uncertainty and instability rather than the expected entropy (version 1). Model instability represented the standard deviation of all predictions for a given input videoclip, and model uncertainty represented 1 minus the maximum prediction class probability. Consistent with version 1, there were decision thresholds incorporated into the categorical reporting which related to prediction class probability (0.5), instability (0.20), and uncertainty (0.44);

- 4) **Suggestive:** Prediction class probability greater than or equal to 0.5, and instability less than 0.20 and uncertainty less than 0.44
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- 6) **Non-diagnostic/Inconclusive:** Instability greater than or equal to 0.20 or uncertainty greater than or equal to 0.44, regardless of prediction class probability.

Page 15, line 12 now reads:

Discrimination performance was overall worse than on internal validation (e.g., AUROC 0.798 vs. 0.95).¹⁷ This degree of performance decrement may be expected when applying an optimistic model to an external cohort.³⁴ Nevertheless, this underlines the need for external validation to identify how such models generalize to real-world clinical practice, particularly as they are further developed.³⁵ Changes to this AI HFpEF model following experience in real-world implementation have yet to be examined,³⁶ and as such, this study therefore represents initial validation experience in a different integrated healthcare delivery network (Mayo Clinic vs. Beth Israel Lahey Health) where care patterns may differ. Despite the observed drop in discrimination, the AI HFpEF model and existing clinical scores demonstrated similar classification characteristics as those previously observed.¹⁷ Importantly, the integration of the AI HFpEF model demonstrated similar utility, highlighting that such iterative improvements to the model appear to have retained overall diagnostic and prognostic performance and generalizability.

2. Much emphasis is placed on the lower rate of ‘non-diagnostic’ results using AI HFpEF versus the clinical scores, but I think these comparisons are a bit artificial at least for H2FPEF. I understand that the H2FPEF publication promoted a score ≥ 6 as high probability for simplicity and as a practical value for targeting high PPV, but since the score translates into a continuous probability it is not as though scores classified here as ‘indeterminate’ (2-5) provide no information. Indeed, we know from the AUROC and curves that both AI HFpEF and H2FPEF actually contain the same amount of predictive information overall. I believe a fairer comparison would be to compare AI HFpEF with H2FPEF when both are thresholded at the same probability value (e.g., 0.5).

This is an important point, which we agree should be addressed appropriately. We agree that the best interpretation of a “non-diagnostic” output does not necessarily mean that there is “no information,” but rather than scores in the middle range should prompt additional testing to further clarify the diagnosis. While model discrimination may be similar between the AI HFpEF and H2FPEF models (as evidenced by the AUROC values), the predictive information may be orthogonal and thus complementary when the AI HFpEF and clinical scores information is combined.

However, we agree that there is a need to better understand how the AI HFpEF model integrates with the clinical scores across the range of possible score values and at similar probability cutoffs. As such, we have instituted several changes to the manuscript accordingly.

First, we now better clarify where classification statistics include all patients vs. only those with diagnostic outputs.

Page 6, line 16 now reads:

A third categorical output, “no classification”, herein referred to as “non-diagnostic,” is generated based on model instability and uncertainty, to ensure that unstable and uncertain predictions are not provided to a user.

Page 7, line 8 now reads:

Using the H2FPEF score, patients were categorized as having low (0 or 1), indeterminate (i.e., “non-diagnostic”; 2-5), or high probability (6-9) of HFpEF. Using the HFA-PEFF score, patients were categorized as unlikely (0-1), indeterminate (i.e., “non-diagnostic”; 2-4), or probable (5-6) HFpEF.

Page 9, line 14 now reads:

*The proportion of cases with normal, abnormal, and missing echocardiographic and clinical parameters is summarized in **Supplemental Figure 1**.*

Page 10, line 7 now reads:

All three clinical scores provided a “non-diagnostic” classification when there was discordant, or missing information to reliably suggest HFpEF presence or absence. Classification performance is therefore presented when considering all patients (i.e., including non-diagnostic results in the calculation of classification statistics), and when considered only diagnostic outputs (i.e., positive or negative).

Page 10, line 14 now reads:

The H2FPEF score resulted in 306 (61.7%) non-diagnostic classifications, compared with 75 (15.1%) for the AI HFpEF model. Compared to the H2FPEF score, the AI HFpEF model demonstrated higher sensitivity (mean difference, 26.7%, 95% CI 17.9 - 35.4%, $p < 0.001$), and specificity (mean difference, 37.5%, 95% CI 29.9 - 45.1%, $p < 0.001$), but no difference in NPV (-8.7%, 95% CI -22.8-5.3%, $p = 0.36$), and PPV (-6.3%, 95% CI -15.7-3.1%, $p = 0.21$). When considering only diagnostic outputs, the H2FPEF score demonstrated higher sensitivity ($p = 0.002$) and specificity ($p = 0.011$).

Owing to the meaningful impact of non-diagnostic classifications on model performance, due either to missing or discordant clinical parameters (detailed in **Supplemental Methods**), classification performance was also assessed only in patients with all information available. After removing patients with missing inputs for the H2FPEF score (n=319 [64.3%]), H2FPEF sensitivity improved (97.5%, 95% CI 92.5-100%, vs. 86.3%, 95% CI 79.8-91.2% for the AI HFpEF model), and specificity improved (38.6%, 95% CI 18.7-53.1% vs. 59.9%, 95% CI 49.2-70.7% in the AI HFpEF model), despite retaining high non-diagnostic rates (61.1% vs. 13.5% in AI HFpEF model; **Supplemental Table 6**).

Page 11, line 9 now reads:

There were 269 (54.2%) non-diagnostic classifications with the HFA-PEFF score. Compared with the HFA-PEFF score, the AI HFpEF model also demonstrated higher sensitivity (by 17.5%, 95% CI 8.8 to 26.2%, $p<0.001$), specificity (by 23.0%, 95% CI 14.5 to 31.6%, $p<0.001$), but lower NPV (by -16.9%, 95% CI -9.0 to -24.8%, $p=0.002$), and PPV (by -19.0%, 95% CI -10.8 to -27.3%, $p<0.001$). However, when considering only diagnostic outputs, the HFA-PEFF score also demonstrated high sensitivity ($p<0.001$) and specificity ($p=0.012$). **Supplemental Table 7** illustrates how performance of the HFA-PEFF score changes as a consequence of the decision threshold. After removing patients with missing inputs for the HFA-PEFF score (n=246 [49.6%]), sensitivity improved to 100% vs. 87.0% (95% CI 81.4-92.1%) in AI HFpEF model, but specificity decreased (26.7%, 95% CI 9.8-42.2% vs. 52.6%, 95% CI 37.3-69.1% for AI HFpEF model), and non-diagnostic rates also decreased (31.3% vs. 14.2% in AI HFpEF model; **Supplemental Table 6**).

Page 13, line 10 now reads:

Similar associations were also present for the clinical scores (**Supplemental Table 10; Supplemental Figures 4-5**), but there was no significant association between the AI HFpEF model and risk of outcomes in non-diagnostic patients according to the H2FPEF and HFA-PEFF scores (**Supplemental Table 12**).

Second, we now provide performance characteristics of the H2FPEF and HFA-PEFF scores at different categorical decision thresholds:

Page 11, line 3 now reads:

Supplemental Table 7 illustrates how performance of the H2FPEF changes as a consequence of the decision threshold.

Page 11, line 14 now reads:

Supplemental Table 7 illustrates how performance of the HFA-PEFF score changes as a consequence of the decision threshold.

Supplemental Table 7 now reads:

Model	Subgroup	Sensitivity (%)	Specificity (%)	NPV (%)	PPV (%)
H2FPEF	>1	98.8 (96.9, 100)	10.9 (6.6, 16.5)	90.3 (75, 100)	51.0 (46.1, 55.8)
	>2	92.5 (87.1, 96.6)	32.8 (26.2, 39.7)	82.4 (72, 91.4)	56.3 (51.2, 60.8)
	>3	79.2 (72.7, 84.6)	57.8 (51.2, 64.3)	74.7 (66.5, 81.9)	63.8 (56.6, 68.9)
	>4	64.6 (58.1, 70.9)	69.9 (63.2, 75.2)	67.8 (61.0, 73.0)	66.8 (59.1, 72.0)
	>5	48.8 (42.3, 56.1)	83.6 (77.8, 88.0)	63.5 (58.3, 68.8)	73.6 (64.9, 79.3)
	>6	22.5 (17.6, 27.6)	94.1 (90.5, 96.8)	56.4 (51.9, 61.8)	78.3 (67.2, 88.0)

	>7	12.5 (8.0, 17.4)	97.7 (95.3, 99.5)	54.3 (49.6, 59.8)	83.3 (70.0, 95.9)
	>8	3.8 (1.1, 6.7)	100 (100, 100)	52.6 (48.1, 57.7)	100 (100, 100)
	>=9	3.8 (1.1, 6.7)	100 (100, 100)	52.6 (48.1, 57.7)	100 (100, 100)
HFA-PEFF	>1	99.6 (98.3, 100)	25.4 (18.1, 30.7)	98.5 (94.7, 100)	55.6 (49.9, 59.8)
	>2	92.1 (87.4, 95.5)	54.3 (46.3, 61.8)	88 (81.1, 93.1)	65.4 (58.8, 70.1)
	>3	84.2 (78.0, 88.9)	69.1 (61.3, 74.6)	82.3 (76.5, 87.6)	71.9 (64.8, 77.1)
	>4	57.9 (51.7, 65.8)	91.4 (87.6, 94.8)	69.9 (64.6, 75.1)	86.3 (79.3, 91.1)
	>5	44.6 (37.6, 52.2)	95.3 (91.9, 98.0)	64.7 (59.8, 67.0)	89.9 (83.5, 95.3)
	>=6	44.6 (37.6, 52.2)	95.3 (91.9, 98.0)	64.7 (59.8, 67.0)	89.9 (83.5, 95.3)

Third, we now compare the AI HFpEF model to the H2FPEF score when categorized at the same decile of continuous probability output:

Page 11, line 4 now reads:

Similarly, **Supplemental Table 8** demonstrates consistent performance between the AI HFpEF model and the H2FPEF score for comparable decision thresholds according to the continuous probability produced by each model.

Supplemental Table 8 now reads:

Continuous Threshold	Model	Sensitivity (%)	Specificity (%)	NPV (%)	PPV (%)
0.10	AI HFpEF	99.2 (97.8, 100)	20.7 (14.5, 26.0)	96.4 (90.5, 100)	54.0 (49.7, 58.3)
	H2FPEF	99.2 (97.9, 100)	7.0 (3.3, 11.3)	90.0 (69.9, 100)	50.0 (44.7, 54.3)
0.20	AI HFpEF	95.4 (92.2, 98.0)	32.8 (26.4, 38.6)	88.4 (80.8, 94.9)	57.1 (51.5, 61.8)
	H2FPEF	97.5 (94.8, 99.7)	17.2 (12.0, 21.2)	88.0 (78.8, 98.9)	52.5 (46.9, 57.5)
0.30	AI HFpEF	92.9 (89.1, 95.9)	41.0 (34.7, 47.0)	86.1 (78.2, 92.3)	59.6 (54.3, 63.5)
	H2FPEF	95.4 (91.3, 97.9)	23.8 (18.5, 29.2)	84.7 (74.5, 93.1)	54.0 (48.7, 59.2)
0.40	AI HFpEF	88.7 (83.0, 92.3)	47.3 (40.1, 53.4)	81.8 (74.1, 88.2)	61.2 (55.7, 65.4)
	H2FPEF	90.8 (85.7, 94.8)	34.4 (27.8, 41.5)	80.0 (71.4, 88.6)	56.5 (50.9, 61.4)
0.50	AI HFpEF	83.8 (78.2, 88.1)	53.9 (47.5, 60.0)	78.0 (70.3, 82.8)	63.0 (56.1, 67.0)
	H2FPEF	85.8 (80.5, 90.6)	43.4 (36.3, 49.9)	76.6 (68.3, 83.7)	58.7 (52.6, 64.3)
0.60	AI HFpEF	77.5 (70.3, 83.3)	59.4 (53.0, 66.2)	73.8 (66.2, 79.5)	64.1 (57.2, 68.1)

	<i>H2FPEF</i>	81.2 (74.6, 86.6)	54.3 (48.0, 61.1)	75.5 (68.5, 81.7)	62.5 (56.8, 67.3)
0.70	<i>AI HFpEF</i>	72.9 (66.0, 78.5)	64.5 (57.8, 70.9)	71.7 (64.8, 77.2)	65.8 (58.9, 70.0)
	<i>H2FPEF</i>	73.8 (66.5, 79.4)	67.2 (61.5, 74.5)	73.2 (66.8, 78.5)	67.8 (61.6, 74.1)
0.80	<i>AI HFpEF</i>	66.2 (58.8, 72.6)	72.7 (67.0, 77.9)	69.7 (62.3, 75.4)	69.4 (61.6, 74.4)
	<i>H2FPEF</i>	64.2 (57.3, 69.5)	78.1 (72.0, 83.2)	69.9 (64.4, 74.0)	73.3 (66.6, 78.6)
0.90	<i>AI HFpEF</i>	57.1 (49.2, 63.8)	81.6 (75.0, 86.0)	67.0 (60.2, 73.2)	74.5 (66.0, 80.4)
	<i>H2FPEF</i>	42.1 (35.0, 47.5)	89.8 (86.1, 93.5)	62.3 (56.5, 66.8)	79.5 (72.3, 85.8)

Fourth, we now remove patients in whom the non-diagnostic output might be due to missing data and compare models:

Page 10, line 22 now reads:

*After removing patients with missing inputs for the H2FPEF score (n=319 [64.3%]), H2FPEF sensitivity improved (97.5%, 95% CI 92.5-100%, vs. 86.3%, 95% CI 79.8-91.2% for the AI HFpEF model), and specificity improved (38.6%, 95% CI 18.7-53.1% vs. 59.9%, 95% CI 49.2-70.7% in the AI HFpEF model), despite retaining high non-diagnostic rates (61.1% vs. 13.5% in AI HFpEF model; **Supplemental Table 6**).*

Page 11, line 15 now reads:

*After removing patients with missing inputs for the HFA-PEFF score (n=246 [49.6%]), sensitivity improved to 100% vs. 87.0% (95% CI 81.4-92.1%) in AI HFpEF model, but specificity decreased (26.7%, 95% CI 9.8-42.2% vs. 52.6%, 95% CI 37.3-69.1% for AI HFpEF model), and non-diagnostic rates also decreased (31.3% vs. 14.2% in AI HFpEF model; **Supplemental Table 6**).*

Fifth, we now estimate reclassification performance parameters to demonstrate that the AI HFpEF model provides meaningful reclassification:

Page 11, line 22 now reads:

Figure 3 illustrates the reclassification of patients from the H2FPEF Score (panel A) and HFA-PEFF Score (panel B) into the AI HFpEF model, alongside classification statistics for the AI HFpEF model for patients otherwise classified as non-diagnostic by these models. For the continuous probability output from the AI HFpEF model and the H2FPEF score, the AI HFpEF model provided significantly more information (likelihood ratio test, $p<0.001$), supported by positive overall net reclassification statistics (NRI 0.40, 95% CI 0.23 to 0.57), and integrated discrimination improvement (IDI) index (0.07, 95% CI 0.01 to 0.13). When comparing the categorical output of the three clinical scores, the AI HFpEF model resulted in a significant increase in information compared with both the H2FPEF (likelihood ratio test, $p<0.001$, NRI: 0.03, 95% CI -0.06 to 0.20) and the HFA-PEFF score (likelihood ratio test, $p<0.001$, NRI: 0.26, 95% CI -0.04 to 0.33; **Figure 3**).

3. More information is needed regarding inclusion criteria and outcomes

a. How was the 2018-2022 window chosen?

Thank you for this comment. The 2018-2022 study window was chosen for several reasons. First, at our institution, our TTE reporting software changed to EncaptureMD in 2018. Structured biplane LVEF data was not captured in the prior iteration of the TTE reporting software, requiring us to limit our study to post-2018. At the time of the analysis, 2022 was the most current year of data available. We also feel that the 2018-2022 study window is reflective of contemporary diagnosis and management of HFpEF which has changed considerably over the past few decades. We have better justified the choice of the study window as follows.

Page 4, line 14 now reads:

This time window was chosen to reflect contemporary management of HFpEF and permit abstraction of structured TTE data on biplane left ventricular ejection fraction (LVEF), not available prior to 2018.

b. Why limit to 300 cases to match (rather than all the potential cases)? How was 300 determined?

We apologize for this important omission which was originally intended to be included in the Supplemental Methods. We have now added relevant information as detailed below.

Page 7, Line 18 now reads:

*The desired sample size was estimated according to modelling of estimated AI HFpEF model performance compared with previously identified clinical benchmarks, similar to previously published,¹⁷ and detailed further in the **Supplemental Methods**.*

Supplemental Methods

Sample Size Estimation

An a priori sample size estimation was completed consistent with the original validation of the AI model (Akerman et al., JACC Adv, 2023). The performance benchmarks generated for the original validation of the model (74% and 65% for sensitivity and specificity, respectively), were used as a minimum acceptable performance. These were tested using a One-sided Binomial Exact Test. The observed sensitivity and specificity from the original validation trial (Akerman et al., JACC Adv, 2023) were used to estimate likely performance in a separate external validation cohort; sensitivity (87.8%; 95% CI: 84.5%-90.9%) and specificity (81.9%; 95% CI: 78.2%-85.6%). We assumed that the lower bound of the confidence interval was an accurate representation of likely performance in a different cohort (i.e., 85% and 78%, respectively), and the lower bound of an anticipated confidence interval approximately 10% lower. Thus, our anticipated performance for sensitivity (lower bound of confidence interval) was 75%, and for specificity was 68%.

We therefore completed a sequence of simulations of One-sided Binomial Exact Tests assuming a sample size range of 100 to 400, an observed sensitivity of 75% to 90%, and observed specificity of 68% to 85%. The upper bound of these simulations was generated from the upper bound of the confidence interval from the previous validation. A full grid of permutations of these values for sensitivity and specificity was generated, resulting in 8,928 possible permutations of sensitivity and specificity performance. For each permutation, two One-sided Binomial Exact Tests compared the relevant sensitivity to the lowest acceptable benchmark (74%), and likewise for specificity (65%). We then compared simulations of significant One-sided Binomial Exact Tests with the feasibility of obtaining cases from the independent site (Beth Israel Deaconess Medical Center, Boston, Massachusetts, USA), and decided that approximately 300 cases and matched 300 controls would permit a maximum of 20% dropout due to uncertain outputs, and still significantly ($p < 0.025$) higher performance than the benchmark for a wide range of observed sensitivity (80%-90%) and specificity (71.2%-85%) permutations.

c. More detail is needed regarding “technically evaluable images”. What does this mean? Note that 20% of individuals in each group were excluded for this vague criterion. It is also curious why this criterion was applied downstream of the case/control matching...why wouldn’t the image quality criterion be applied early on so that 300 cases and controls could be robustly secured?

Thank you for this comment. Technically evaluable images were those for which the AI HFpEF model was able to generate an output. Largely this reflects the 10-20% of individuals who technically difficult echocardiographic windows (Fraiche, *J Am Soc Echocardiogr*, 2020) defined as 2 or more nonvisible endocardial segments. We did not match cases and controls on image quality due to concerns about bias. As the presence of obesity is related both to image quality (Fraiche, *J Am Soc Echocardiogr*, 2020) as well as the development of HFpEF (Borlaug, *Cardiovasc Res*, 2022), matching on image quality could introduce selection bias in that matching would select cases with overall improved image quality compared to the average patient with HFpEF (or controls that have worse image quality). However, to the extent that this exclusion limits diagnostic evaluation using the AI HFpEF model, we would consider the results of our analysis to be conditional on having adequate quality imaging. This has been added as a limitation as follows.

Page 17, line 10 now reads:

Second, categorical outputs from all three models incorporate a non-diagnostic classification intended to support further confirmatory testing; direct comparison between models therefore must consider whether non-diagnostic outputs are due to missingness or discordance, and whether this would occur in clinical implementation. For instance, NT-proBNP was missing in nearly a quarter of patients, reflecting the challenges of guideline adherence in clinical practice as well as the clinical scores that utilize such laboratory markers, but could be obtained as required. When non-diagnostic outputs were considered a relevant element in the decision-making process, the current study demonstrated superiority of the AI HFpEF model. However, it must also be acknowledged that several patients were excluded due to poor image quality. While this degree of poor image quality is within published norms for echocardiography,⁴¹ it nevertheless exerts a greater effect on the AI HFpEF model compared to multiparametric clinical scores.

d. Almost half of the TTEs were excluded for missing biplane LVEF. This seems to be a large proportion. Is biplane LVEF measurement standard at this institution? If not, why was it chosen as a selection criterion? Are there temporal/trends in practice (or other potential indirect mechanisms of selection) that need to be considered here?

Thank you for this insightful comment. Of those missing biplane LVEF, nearly half have a 3D quantitative LVEF value and the other half have a visually-estimated LVEF. This pattern was consistent across the years of study. While one could argue that 3D quantitative LVEF should suffice to establish the presence of a preserved LVEF, we chose to focus the analysis on only those with biplane LVEF given 1) possible differences between LVEF techniques, 2) use of biplane LVEF to define HFpEF in the derivation manuscript, 3) the fact that American Society of Echocardiography guidelines continue to advocate for use of biplane LVEF (Lang et al. *J Am Soc Echocardiogr*, 2015). We have further justified this choice in the manuscript as follows.

Page 4, line 18 now reads:

The focus on biplane LVEF was for several reasons: 1) there may be possible differences across LVEF techniques,¹⁸ 2) current American Society of Echocardiography guidelines advocate for use of biplane LVEF,¹⁸ and 3) the criteria for ejection fraction in the development of the AI HFpEF model used biplane LVEF values.¹⁷

e. Controls had “no evidence of HF within 1-year prior to and after the index TTE”. Does this mean no

inpatient HF code in this timeframe? Or any HF code? Or any evidence of HF as determined by the manual chart review?

Thank you for this important point of clarification. We apologize for not being clear previously. Controls were selected if they lacked an inpatient HF hospitalization code in the primary position in the year prior to or following the index TTE date. Secondly, we adjudicated whether HF was truly absent in the year prior and after the date of TTE on manual chart review, evaluating both inpatient and outpatient medical records. However, this adjudication was not used to select controls but rather verify that the selection of controls was appropriate. On chart review, no controls were found to have HF within the year prior to or following the index TTE. We have clarified this further in the manuscript as follows.

Page 5, line 3 now reads:

Additionally, cases were required to have evidence of recent hospital admission for heart failure (HF) within a year prior to the index TTE date, as evidenced by a primary discharge diagnosis of 428.X (under the International Classification of Diseases, 9th Revision [ICD-9] classification) prior to October 1, 2015 or I50.X (under the International Classification of Diseases, 10th Revision [ICD-10] classification) on or after October 1, 2015. Controls were similarly identified as having grade I, or no diastolic dysfunction on TTE, and no evidence of a hospital admission for HF within one-year prior to and after the index TTE date. Presence or absence of HF hospitalization was additionally adjudicated and confirmed at the time of manual chart review, but this information was not used in case or control selection.

Page 9, line 2 now reads:

*A final sample of 240 cases and 256 were identified after application of exclusions (**Figure 1**). All cases had HF within the prior year on manual chart review and all controls lacked HF in the year prior to or following the index TTE.*

f. “Presence or absence of HF hospitalization was additionally adjudicated at the time of manual chart review”. Statement is vague. Were all cases and controls manually reviewed? What criteria were applied to define HF status using manual chart review (both for the purposes of defining HF case status and the HF outcome)?

Thank you. We apologize for the lack of clarity. All cases and controls were manually reviewed to abstract information on outcomes and covariates, but selection for the cohort was based on claims. Manual chart adjudication of HF status was based on presence of any admission for heart failure or chart designation of presence of heart failure. We have clarified this further in the manuscript as follows.

Page 5, line 3 now reads:

Additionally, cases were required to have evidence of recent hospital admission for heart failure (HF) within a year prior to the index TTE date, as evidenced by a primary discharge diagnosis of 428.X (under the International Classification of Diseases, 9th Revision [ICD-9] classification) prior to October 1, 2015 or I50.X (under the International Classification of Diseases, 10th Revision [ICD-10] classification) on or after October 1, 2015. Controls were similarly identified as having grade I, or no diastolic dysfunction on TTE, and no evidence of a hospital admission for HF within one-year prior to and after the index TTE date. Presence or absence of HF hospitalization was additionally adjudicated and confirmed at the time of manual chart review, but this information was not used in case or control selection.

Page 9, line 2 now reads:

*A final sample of 240 cases and 256 were identified after application of exclusions (**Figure 1**). All cases had HF within the prior year on manual chart review and all controls lacked HF in the year prior to or following the index TTE.*

g. Were the adjudicators blinded to HFpEF case/control status when ascertaining clinical variables and outcomes?

Thank you for this comment. As the adjudicators thoroughly reviewed charts including discharge summaries that contained details on presence or absence of HF, it was not possible to blind adjudicators to case or control status. However, the selection of cases and controls was determined prior to chart review so as to avoid information bias. However, it is possible that the covariate or outcome abstraction could have been influenced by lack of blinding which we have now commented on specifically in the methodology.

Page 5, line 22 now reads:

Outcomes included all-cause mortality and HF hospitalization within the year following the index TTE, with occurrence of an outcome abstracted by medical chart review. Patient status (case or control) was not blinded to analysts during extraction of relevant clinical parameters and patient outcomes.

4. The confidence intervals for AUROC (e.g., AI HFpEF 0.798, 95% CI 0.756-0.799) seem off. I understand AUROC CIs can be asymmetric, but the upper bound being 0.799 when the effect estimate is 0.798 and the sample size is modest seems questionable. Likewise for H2FpEF (AUROC 0.788, 95% 0.745-0.798).

This is indeed correct. We appreciate the astute reviewer for identifying this error. There was unfortunately an error in the extraction of the upper bound of the confidence interval for this specific test alone. It should instead be:

AI HFpEF: 0.798 (95% CI: 0.756-0.842)

H2FpEF: 0.788 (95% CI: 0.745-0.831)

This has been changed where relevant throughout the manuscript.

5. BNP/NTproBNP appears to be missing for >25% of HFpEF cases. This seems unexpected for patients with HF as a primary diagnosis for hospitalization. Could the authors comment?

Thank you for this comment. We agree that a number of NT-proBNP values are missing on patients with HFpEF. There may be several reasons for this: 1) cases were not required to be hypervolemic at the time of their index TTE but rather to have a recent (< 1 year) HF exacerbation. Thus, it is possible that NT-proBNP values were drawn at a time when the patient was not in acute HF; 2) Similarly, the most recent NT-proBNP value within 3 months of the index TTE was used and thus could be reflective of a state of relative euvolemia; 3) Given that NT-proBNP values may be low in individuals with obesity despite clinical HF and the association of HFpEF and obesity, it is possible that a portion of these values are falsely negative; 4) the lower than universal rate of NT-proBNP testing may be reflective of guideline nonadherence. We have added this as a limitation and broader contextualisation to the analysis as follows.

Page 17, line 10 now reads:

Second, categorical outputs from all three models incorporate a non-diagnostic classification intended to support further confirmatory testing; direct comparison between models therefore must consider whether non-diagnostic outputs are due to missingness or discordance, and whether this would occur in clinical implementation. For instance, NT-proBNP was missing in nearly a quarter of patients, reflecting the challenges of guideline adherence in clinical practice as well as the clinical scores that utilize such laboratory markers, but could be obtained as required.

6. In the discussion, the authors envision a clinical use case in which AI HFpEF is applied for individuals with indeterminate clinical risk scores. However, I believe the converse (applying clinical risk scores if/only if AI HFpEF is indeterminate) is more practical, as it minimizes the number of individuals who would require complex clinical score availability/calculation, which the authors argue is an important goal.

Thank you for this suggestion. First, it must be made clear that the joint approach in the decision curve analysis relates both to the use case of AI HFpEF first or H2FPEF first. For the conditional approach, the original coding was such that either the existing clinical score was positive or the existing clinical score was indeterminate, and the AI HFpEF model was positive. We opted for the original approach in order to provide a conservative decision-making process (i.e., clinicians using something that is already familiar to them), but to address the reviewer's point regarding the alternative approach, we have now added modelling for when the AI HFpEF model is positive or the AI HFpEF model is indeterminate and the existing clinical score is positive. Both versions are presented in **Supplemental Table 9** (reproduced below for ease).

Supplemental Table 9 now reads:

<i>Clinical Comparison</i>	<i>Decision Making Approach</i>	<i>Net Benefit</i>	<i>Standardized Net Benefit</i>	<i>Net Interventions Avoided</i>
HFA-PEFF	<i>Treat All</i>	0.000	0.000	0.000
	<i>Treat None</i>	0.000	0.000	0.000
	<i>HFA-PEFF</i>	0.148	0.493	0.345
	<i>Either HFA-PEFF or AI HFpEF</i>	0.144	0.480	0.336
	<i>Combined (Clinical Model First)</i>	0.150	0.499	0.349
	<i>Combined (AI HFpEF Model First)</i>	0.141	0.469	0.328
H2FPEF	<i>Treat All</i>	0.000	0.000	0.000
	<i>Treat None</i>	0.000	0.000	0.000
	<i>H2FPEF</i>	0.097	0.323	0.226
	<i>Either H2FPEF or AI HFpEF</i>	0.130	0.434	0.304
	<i>Combined (Clinical Model First)</i>	0.129	0.429	0.301
	<i>Combined (AI HFpEF Model First)</i>	0.135	0.450	0.315
Combined Modelling	<i>Treat All</i>	0.000	0.000	0.000
	<i>Treat None</i>	0.000	0.000	0.000
	<i>H2FPEFF</i>	0.058	0.192	0.135
	<i>AI HFpEF model</i>	0.102	0.339	0.238
	<i>H2FPEFF & HFA-PEFF</i>	0.152	0.506	0.354
	<i>AI-HFpEF model & Clinical models</i>	0.164	0.546	0.382

7. Limitations, “although biplane LVEF values were not verified...AI HFpEF computed biplane LVEFs”. This is the first mention of this process that I can see. How were these computed values utilized in the selection process?

Thank you for allowing us to clarify this point. The biplane LVEFs used for selection into the current study were those reported by faculty echocardiographers in the final study report. While the AI HFpEF model produces a biplane LVEF measurement which could be used for verification that the initial LVEFs were

accurate, these machine-generated LVEF values were not used in patient selection. To avoid confusion, we have modified the language as follows.

Page 5, line 19 now reads:

Biplane left ventricular ejection fraction values were not verified by manual inspection of images, but instead, values from the final study report (by experienced faculty echocardiographers) were extracted, and thus reflect the subsequent patient management pathway.

8. I believe the Discussion would be improved by better focus on the specific use case of AI HFpEF and where the specific added value here. Several points regarding the challenges of HFpEF (absence of a standard definition, heterogeneity of etiology) that are discussed are not addressed by AI HFpEF.

Thank you. We have attempted to provide a more concise and focused narrative to the discussion, and associated analyses, regarding the specific use case we feel would be addressed by the AI HFpEF model.

Page 15, Line 23 now reads:

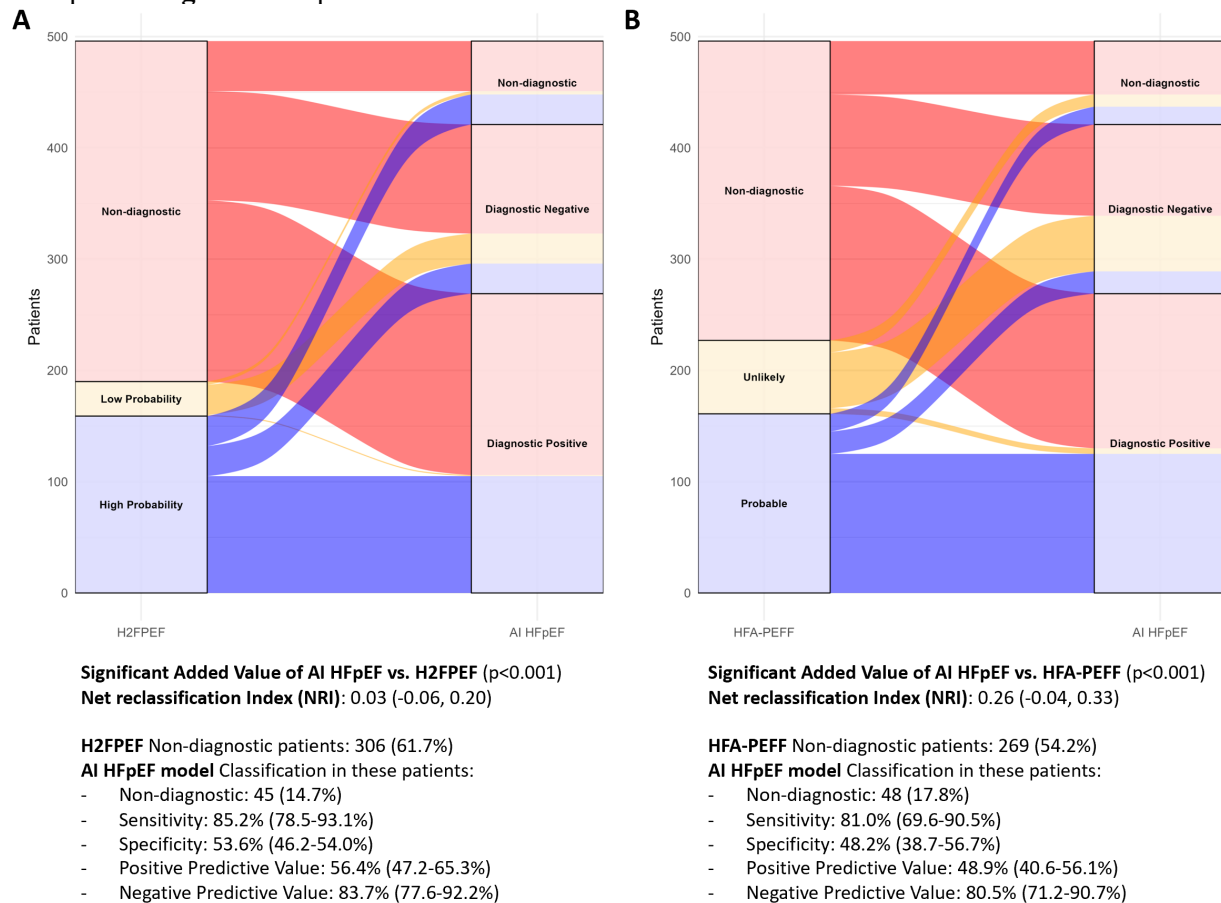
Suggestions to integrate AI models into existing diagnostic pathways^{16,17} and utilize the combined clinical information from all available sources are supported by the current study. In decision modelling, use of combined information from the AI HFpEF model and clinical scores resulted in a greater number of individuals appropriately receiving treatment (in this case SGLT2 inhibitors) with fewer number of individuals treated unnecessarily. This is logical, given that each model is associated with its own development process (AI, statistical modelling, and clinical consensus for AI HFpEF, H2FPEF, and HFA-PEFF, respectively); in a heterogeneous syndrome like HFpEF, each model would likely capture different types of patients under the broad umbrella of HFpEF. This feature becomes especially important given the absence of one single standard definition of HFpEF, and heterogeneity of the etiology of the syndrome; different diagnostic models which can be readily integrated would capitalize on such ambiguity and provide the greatest value for patients. While the AI HFpEF model added significant value to the information contained in the existing clinical scores, the greatest benefit was observed when all models were utilized for identification of disease and decision-making processes. This approach is supported by findings in other disease states such as prediction of atrial fibrillation³⁷ and coronary artery disease risk,³⁸ where the use of AI has been found to be complementary to clinical information. However, the implementation of AI in a supportive role for clinical diagnostics is still in its infancy, and requires prospective testing to understand where and how this integration might provide value.

9. I don't find the Alluvial plots in Figure 3 to be particularly useful. They summarize reclassification but (unless I am misinterpreting them) they do not indicate whether reclassification is appropriate or inappropriate. I believe a categorical net reclassification analysis would be more customary (where non-diagnostic could be considered an intermediate class), and the corresponding tables would give a better sense of whether the reclassification is correct or not. Along the same lines I think presentation of continuous NRI or integrated discrimination improvement (IDI) using the continuous probabilities (i.e., AI HFpEF and H2FPEF) would be useful information.

We agree that the alluvial plots only provide a descriptive summary of the reclassification between models, rather than the appropriateness. This was intentional to demonstrate how the two models differ in their classification. We also agree that a more appropriate test of reclassification should be provided to give an indication of the appropriateness of reclassification and added value of each model.

We have provided the NRI and IDI as suggested (**Figure 3**). Additionally, we have also included a log-likelihood based test (likelihood ratio test) comparing the two models. The NRI and Likelihood ratio test are presented for the continuous and categorical formations of the models, and the IDI for the continuous only.

The updated **Figure 3** is reproduced below:



Page 11, Line 22 now reads:

Figure 3 illustrates the reclassification of patients from the H2FPEF Score (panel A) and HFA-PEFF Score (panel B) into the AI HFpEF model, alongside classification statistics for the AI HFpEF model for patients otherwise classified as non-diagnostic by these models. For the continuous probability output from the AI HFpEF model and the H2FPEF score, the AI HFpEF model provided significantly more information (likelihood ratio test, $p < 0.001$), supported by positive overall net reclassification statistics (NRI 0.40, 95% CI 0.23 to 0.57), and integrated discrimination improvement (IDI) index (0.07, 95% CI 0.01 to 0.13). When comparing the categorical output of the three clinical scores, the AI HFpEF model resulted in a significant increase in information compared with both the H2FPEF (likelihood ratio test, $p < 0.001$, NRI: 0.03, 95% CI -0.06 to 0.20) and the HFA-PEFF score (likelihood ratio test, $p < 0.001$, NRI: 0.26, 95% CI -0.04 to 0.33; **Figure 3**).

10. Some comments regarding the decision curve analysis

a. More information is needed in the caption as many readers may not be familiar with this type of analysis. The information in the Supplemental Methods is useful but much of it should be incorporated in the caption or elsewhere in the main Methods.

Thank you for this comment. We agree and have replaced the caption for **Figure 4** with the information in the **Supplemental Methods**.

The **Figure 4** legend now reads:

LEGEND: Decision curves comparing the standardized net benefit (panel A) and net reduction in interventions (panel B) when patient management decisions are based on the output of a clinically validated models (H2FPEF score) and/or the AI HFpEF model. In the scenarios presented, the management/treatment decision is assumed to represent prescription of SGLT2i to the patient, in a population where the expected prevalence of HFpEF is 30%. The red and gold lines represent clinical baselines of prescribing all patients (red) or no patients (gold), regardless of the output of any test. Prescribing based on only the clinical model classification is represented in green. In this scenario, any patient with a probable classification of HFpEF according to the clinical model would be prescribed SGLT2i. The blue line represents a joint approach, prescribing patients based on either a probable classification from the clinical model, or a positive diagnostic classification from the AI HFpEF model. The purple line represents a conditional approach, wherein patients are prescribed based on either a positive diagnostic classification from the AI HFpEF model, or a non-diagnostic classification from the AI HFpEF model, but a probable classification from the clinical model. The alternative order (clinical model first) is presented in **Supplemental Table 9**. The x-axis refers to the threshold probability that would be required by a clinician and/or patient to initiate prescription of SGLT2i (e.g., a 30% probability threshold means that one would want a patient to have 30% likelihood of having HFpEF before initiating a given management option). This threshold probability likely differs in various clinical scenarios, and for different management options, but generally and simply, it can represent the relative harm of avoiding a given management option for a patient with the disease, compared to unnecessarily intervening on a patient who is disease free. In the context of the current study, the minimum threshold probability one might expect would be 30% (dotted line). This represents the trade-off between the risk of an adverse event when taking SGLT2i (5.8%⁴²), and the risk reduction associated with taking an SGLT2i (-19%⁴² for HF hospitalization or worsening HF event). The x-axis is truncated to clinically reasonable threshold probabilities for clear and meaningful interpretation (i.e., likely unreasonable that a clinician would require a patient to have higher than 40% probability of HFpEF before initiating prescription of SGLT2i). For net benefit plots, the y-axis refers to the standardized net benefit of taking a given approach, with units presenting the proportions of patients with disease in the population who would be successfully managed according to the different approaches. For example, a value of 0.45 for the Combined approach would be interpreted such that, compared to prescribing no patients with SGLT2i, managing patients based on the combined information from the AI HFpEF model and the H2FPEF score would result in 45% of patients with HFpEF being correctly managed. For net reduction in interventions, a value of 0.315 for the Combined approach would be interpreted such that, prescribing SGLT2i based on the combined information would lead to an absolute 31.5% reduction in the number of prescriptions without missing any patients with HFpEF. Absolute values for each decision-making approach at a 30% threshold probability are provided in **Supplemental Table 9**.

b. As currently parameterized, I believe the “Either” and “Combined” approaches are identical. This is supported by the fact the curves are entirely overlapping. That said, as per my comment above I would consider adding a condition where one runs the clinical score only if AI HFpEF is indeterminate.

As detailed above, the alternative approach has now been added. Both versions are presented in **Supplemental Table 9** (reproduced below for ease).

Supplemental Table 9 now reads:

Clinical Comparison	Decision Making Approach	Net Benefit	Standardized Net Benefit	Net Interventions Avoided
---------------------	--------------------------	-------------	--------------------------	---------------------------

HFA-PEFF	<i>Treat All</i>	0.000	0.000	0.000
	<i>Treat None</i>	0.000	0.000	0.000
	<i>HFA-PEFF</i>	0.148	0.493	0.345
	<i>Either HFA-PEFF or AI HFpEF</i>	0.144	0.480	0.336
	<i>Combined (Clinical Model First)</i>	0.150	0.499	0.349
	<i>Combined (AI HFpEF Model First)</i>	0.141	0.469	0.328
H2FPEF	<i>Treat All</i>	0.000	0.000	0.000
	<i>Treat None</i>	0.000	0.000	0.000
	<i>H2FPEF</i>	0.097	0.323	0.226
	<i>Either H2FPEF or AI HFpEF</i>	0.130	0.434	0.304
	<i>Combined (Clinical Model First)</i>	0.129	0.429	0.301
	<i>Combined (AI HFpEF Model First)</i>	0.135	0.450	0.315
Combined Modelling	<i>Treat All</i>	0.000	0.000	0.000
	<i>Treat None</i>	0.000	0.000	0.000
	<i>H2FPEFF</i>	0.058	0.192	0.135
	<i>AI HFpEF model</i>	0.102	0.339	0.238
	<i>H2FPEFF & HFA-PEFF</i>	0.152	0.506	0.354
	<i>AI-HFpEF model & Clinical models</i>	0.164	0.546	0.382

c. I would recommend the x-axis is expanded to the point where the risk stratification approaches intersect with “Treat None” – as this depicts the threshold where risk stratification is no longer useful (a useful data point to show).

We appreciate the hesitancy to truncate the x-axis to be truncated. However, we believe that the context of the current analysis (decision curves) continues to warrant this approach. We have explained our reasoning (with references) below.

Motivating this truncation, the decision curve is meant to inform a very specific decision in the clinical decision-making process; the range of probabilities should therefore be informed by the decision in question.

The following references suggest that the range of threshold probabilities should be accordingly narrowed:

See: <https://diagnprognres.biomedcentral.com/articles/10.1186/s41512-019-0064-7>

And: <https://www.fharrell.com/post/edca/>:

- **“Error 2: Showing too wide a range of threshold probabilities”**
- “It is not uncommon to see a decision curve analysis where the model is clearly intended to inform a specific decision, such as biopsy or prophylactic therapy, but the x axis includes a very wide range of threshold probabilities. Many of these threshold probabilities will be uninformative. For instance, there is no need to know the net benefit for a cancer prediction model at a threshold probability of 80%, because no reasonable decision-maker should demand an 80% risk of cancer before agreeing to biopsy. Unless they are evaluating a prognostic model used for general patient counseling, as described above, investigators should prespecify a limited range of reasonable threshold probabilities and limit the x axis to that range. A reasonable exception to this rule is if the lower bound of the threshold probability is low, such as 5% or 10%, in which case investigators may choose to start the x axis at 0.”

If the reviewer or handling editor believes that this is not sufficiently justified, we would be happy to extend the x-axis, but given that this may confuse interpretation (as detailed in the referenced material), we have left the x-axis unchanged in the current version of the manuscript.

d. Since the authors propose a risk threshold based on the DELIVER trial, I think it would be valuable to report the actual net benefit values and net interventions avoided for each strategy at that threshold.

Thank you for highlighting this. We agree and apologize for the omission from the original Supplemental Appendix. Dependent on which outcome you choose from DELIVER, HF hospitalization/worsening HF event or composite of worsening HF or CV death, you end up with different thresholds (30% and 36%, respectively). The identified threshold on the figures relates to the most conservative estimate (30%). The actual values at this threshold are now provided in **Supplemental Table 9**, which is provided below.

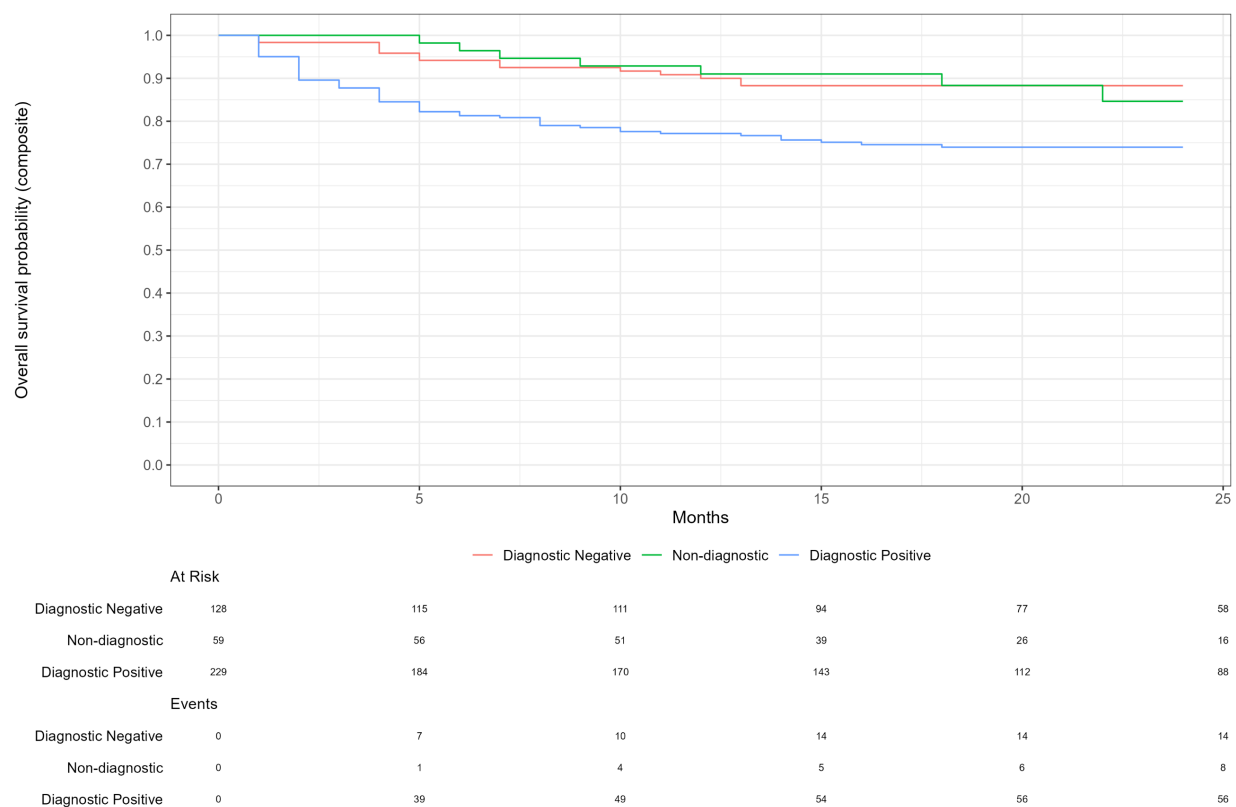
Supplemental Table 9 now reads:

<i>Clinical Comparison</i>	<i>Decision Making Approach</i>	<i>Net Benefit</i>	<i>Standardized Net Benefit</i>	<i>Net Interventions Avoided</i>
HFA-PEFF	<i>Treat All</i>	<i>0.000</i>	<i>0.000</i>	<i>0.000</i>
	<i>Treat None</i>	<i>0.000</i>	<i>0.000</i>	<i>0.000</i>
	<i>HFA-PEFF</i>	<i>0.148</i>	<i>0.493</i>	<i>0.345</i>
	<i>Either HFA-PEFF or AI HFpEF</i>	<i>0.144</i>	<i>0.480</i>	<i>0.336</i>
	<i>Combined (Clinical Model First)</i>	<i>0.150</i>	<i>0.499</i>	<i>0.349</i>
	<i>Combined (AI HFpEF Model First)</i>	<i>0.141</i>	<i>0.469</i>	<i>0.328</i>
H2FPEF	<i>Treat All</i>	<i>0.000</i>	<i>0.000</i>	<i>0.000</i>
	<i>Treat None</i>	<i>0.000</i>	<i>0.000</i>	<i>0.000</i>
	<i>H2FPEF</i>	<i>0.097</i>	<i>0.323</i>	<i>0.226</i>
	<i>Either H2FPEF or AI HFpEF</i>	<i>0.130</i>	<i>0.434</i>	<i>0.304</i>
	<i>Combined (Clinical Model First)</i>	<i>0.129</i>	<i>0.429</i>	<i>0.301</i>
	<i>Combined (AI HFpEF Model First)</i>	<i>0.135</i>	<i>0.450</i>	<i>0.315</i>
Combined Modelling	<i>Treat All</i>	<i>0.000</i>	<i>0.000</i>	<i>0.000</i>
	<i>Treat None</i>	<i>0.000</i>	<i>0.000</i>	<i>0.000</i>
	<i>H2FPEFF</i>	<i>0.058</i>	<i>0.192</i>	<i>0.135</i>
	<i>AI HFpEF model</i>	<i>0.102</i>	<i>0.339</i>	<i>0.238</i>
	<i>H2FPEFF & HFA-PEFF</i>	<i>0.152</i>	<i>0.506</i>	<i>0.354</i>
	<i>AI-HFpEF model & Clinical models</i>	<i>0.164</i>	<i>0.546</i>	<i>0.382</i>

11. Figure 5 (KM curves) – Why do the curves not start at 1?

Thank you for highlighting this. The KM curves not starting at 1 for hospitalization outcomes is explained by the time difference between the hospitalization date and the echocardiogram date. Specifically, there were 12 patients who were hospitalized prior to their echocardiogram. Given that this hospitalization might have formed a part of the patient selection process, and thus would introduce potential bias, we have removed these patients from the analysis and updated all relevant figures and tables in the main document and supplemental appendix.

The updated **Figure 5** now is reproduced below:



12. The AI HFpEF model appears to display an unexpected behavior such that that true risk appears to decrease from quartile 2 to quartile 3. This is visible in the calibration plots (where quartile 3 is particularly miscalibrated) as well as Supplemental Table 7 where the HRs for quartile 3 versus quartile 1 are consistently lower than those for quartile 2 versus quartile 1. This seems to suggest an issue with model predictions in this range and requires further discussion/explanation.

Thank you for this astute comment. We agree and see this as suggesting that the model predicts less well in the mid-range probabilities. We have added this to the manuscript as

Page 9, line 22 now reads:

Both the AI HFpEF model and H2FPEF score demonstrated similar calibration results. The intercept and slope for the AI HFpEF model (intercept: -0.56, 95% CI -0.82 to -0.32; slope: 0.44, 95% CI 0.35 to 0.56) and H2FPEF score (intercept: -1.00, 95% CI -1.75 to -0.40; slope: 0.81, 95% CI 0.58 to 1.15) both indicated an overestimation of risk. For the AI HFpEF model this overestimation was predominant in moderate-to-high probabilities, whereas for the H2FPEF score there was overestimation across the full spectrum of probabilities (Supplemental Figure 2, Hosmer-Lemeshow, both $p < 0.001$).

1. In general there is too much discussion regarding the nuances of WPW, which is a rare condition, particularly when compared to the other abnormalities tested (RBBB, LBBB, etc).

Thank you for the comment. Unfortunately, we are unable to find any references to WPW or conduction abnormalities (e.g. RBBB or LBBB) in the manuscript, which does not mention electrocardiography. We apologize that we cannot respond accordingly but would be happy to readdress this point if the reviewer is able to indicate the page and lines in which these are mentioned.

2. Results, line 236 – not clear what the unit of these effect sizes are. Is it per-quartile increase?

Thank you for this comment. We apologize for the lack of initial clarity. These effect sizes represent hazard ratios as a comparison to the reference group (i.e., quartile 1). We have now made this clear.

Page 13, line 7 now reads:

Compared to the first quartile, risk of the composite outcome (HR 3.95, 95% CI 2.00-7.81, $p<0.001$), all-cause mortality (HR 3.14, 95% CI 1.40-7.02, $p=0.005$), and HF hospitalization (HR 4.74, 95% CI 1.61-13.9, $p=0.005$) were all elevated (Supplemental Table 11).

3. Discussion, line 314 and elsewhere in the paragraph – patients were not really ‘enrolled’ in this retrospective study. Would rephrase.

Thank you. We have rephrased this throughout the manuscript.

Page 3, line 24 now reads:

As such, in this study, we aimed to evaluate the performance of a novel AI HFpEF model (EchoGo Heart Failure, Ultromics Ltd) in an independent dataset, and compare this to existing multiparametric clinical scores for diagnosis of HFpEF in a matched case control study including patients with and without HFpEF who underwent TTE at Beth Israel Deaconess Medical Center (BIDMC).

Page 6, line 18 now reads:

In the initial derivation study which included 2,971 cases and 3,785 controls, the initial version of the AI HFpEF model demonstrated excellent discrimination for both training (area under the receiver operating characteristic curve [AUROC], 0.97, 95% CI 0.96-0.97) and validation (AUROC 0.95, 95% CI 0.93-0.96) data, which was maintained in independent testing (AUROC 0.91, 95% CI 0.90-0.93), and identified individuals at greater risk of mortality.¹⁷

Page 17, line 1 now reads:

While including a rigorously phenotyped population with and without HFpEF, the current study has limitations worth acknowledgement. First, it must be acknowledged that the clinical HFpEF syndrome has no clear and consistent definition, and a heterogenous etiology, meaning that not all HFpEF phenotypes and definitions will be captured in the current study.

4. Conclusions, line 330 – “higher accuracy than clinical scoring systems to classify patients...due to non-diagnostic rates from clinical scores”. I do not think the current results support such a statement from an “accuracy” standpoint. Would rephrase.

Thank you for this comment. We agree that the results do not support this statement when non-diagnostic classifications are removed. Therefore, we have removed references to accuracy throughout the manuscript and have taken great care to contextualize the results throughout the results and discussion.

Page 10, line 5 now reads:

Table 2 highlights classification performance for all models. Full confusion matrices are provided in Supplemental Table 4 and comparison of performance across sociodemographic groups in Supplemental Table 5. All three clinical scores provided a “non-diagnostic” classification when there was discordant, or missing information to reliably suggest HFpEF presence or absence. Classification performance is therefore presented when considering all patients (i.e., including non-diagnostic results in the calculation of classification statistics), and when considered only diagnostic outputs (i.e., positive or negative).

Page 17, line 10 now reads:

Second, categorical outputs from all three models incorporate a non-diagnostic classification intended to support further confirmatory testing; direct comparison between models therefore must consider whether non-diagnostic outputs are due to missingness or discordance, and whether this would occur in clinical implementation. For instance, NT-proBNP was missing in nearly a quarter of patients, reflecting the challenges of guideline adherence in clinical practice as well as the clinical scores that utilize such laboratory markers, but could be obtained as required. When non-diagnostic outputs were considered a relevant element in the decision-making process, the current study demonstrated superiority of the AI HFpEF model. However, it must also be acknowledged that several patients were excluded due to poor image quality. While this degree of poor image quality is within published norms for echocardiography,⁴¹ it nevertheless exerts a greater effect on the AI HFpEF model compared to multiparametric clinical scores.

Page 18, line 2 now reads:

In this case control study, an AI HFpEF model using deep learning of a single TTE video clip had higher classification performance than existing clinical scores, largely owing to high rates of non-diagnostic outputs from multivariable scores comprised of discordant clinical parameters. The AI HFpEF model demonstrated added value to the diagnostic process, but the greatest benefit was observed when information from existing clinical scores and novel AI are integrated into the decision-making process.

5. Discussion points regarding complementarity of AI and clinical information would be improved by referencing related work suggesting similar conclusions in areas such AF prediction (e.g., PMID 34743566) and CAD risk estimation (e.g., PMID 37700032) using ECG.

Thank you for highlighting these references which support the complementarity of AI and clinical information. We have updated the manuscript accordingly.

Page 16, line 8 now reads:

While the AI HFpEF model added significant value to the information contained in the existing clinical scores, the greatest benefit was observed when all models were utilized for identification of disease and decision-making processes. This approach is supported by findings in other disease states such as prediction of atrial fibrillation³⁷ and coronary artery disease risk,³⁸ where the use of AI has been found to be complementary to clinical information.

The authors would like to thank the reviewers for these important comments which we feel have strengthened the manuscript. We now hope that the manuscript is suitable for publication in *Nature Communications*.

Response to Reviewer 1:

We would like to thank the reviewer for their thoughtful comments on our paper which we believe have strengthened the manuscript.

I thank the authors for responding to my previous comments. I have only few comments remaining:

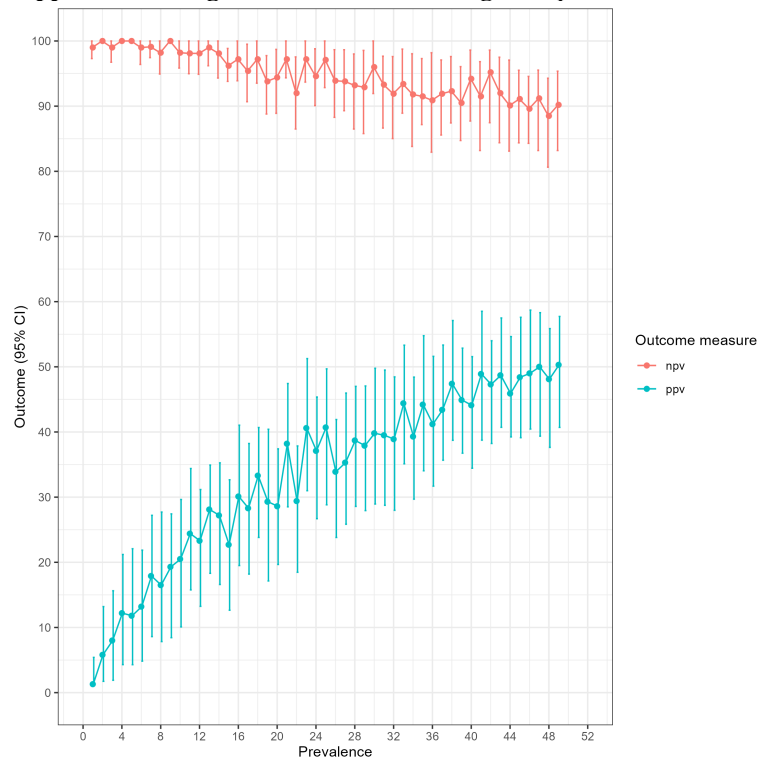
NPV and PPV are dependent on the prevalence in the study population, which here is up-sampled due to the case-control design – as far as I understand these numbers were not adjusted for expected real-world prevalence, which hinders direct translation of NPV/PPV to the clinical setting and should be clarified for the reader.

Thank you, this had not been highlighted. We have added this contextual information to the manuscript, and also added another analysis to the supplement to detail the performance with resampled data at different prevalences.

Page 8, line 4 now reads:

*Classification performance, including the sensitivity, specificity, negative predictive value (NPV), and positive predictive value (PPV) was determined for all models (**Supplemental Table 3**). Proportions tests are used for statistical comparisons between models for classification performance. Predictive values are reported according to the prevalence in the case-control design, and thus are expected to be higher than might be encountered in clinical practice; NPV and PPV are therefore reported at alternative prevalence(s), according to resampling of the current data, in **Supplemental Figure 2**.*

Supplemental Figure 2. Positive and negative predictive value as a function of disease prevalence



LEGEND: Displayed are positive (blue) and negative (red) predictive value (PPV and NPV, respectively) as a function of prevalence (x-axis; %), when the dataset is resampled to achieve a given prevalence. Specifically, controls are resampled to 80% of the original total, and cases are randomly sampled to achieve

prevalence ranging from 1% to the case-control prevalence (52%). Bootstrap 95% confidence interval are provided alongside point estimates.

Abstract: “Models with continuous probability outputs (AI and H2FPEF) demonstrated similar discrimination and calibration, but the AI HFpEF model had higher sensitivity (AI: 77.4% vs. H2FPEF: 53.9% vs. HFA-PEFF: 63.2%) and specificity (50.2% vs. 12.8% vs. 29.0%), largely attributable to higher non-diagnostic rates (15.1% vs. 61.7% vs. 54.2%), even after accounting for missing multivariable inputs.” Should this not be *lower* non-diagnostic rates, as it refers to the AI HFpEF model?

Thank you for highlighting this error, it has now been fixed.

The abstract now reads:

Artificial intelligence (AI) models to identify heart failure (HF) with preserved ejection fraction (HFpEF) based on deep-learning of echocardiograms could help address under-recognition in clinical practice, but they require extensive validation, particularly in representative and complex clinical cohorts for which they could provide most value. Patients with HFpEF (cases; n=240), and age, sex, and year of echocardiogram matched controls (n=256) were identified from clinically-indicated clinical examinations. Diagnostic performance (discrimination, calibration, classification, and clinical utility) and prognostic associations (mortality and HF hospitalization) were compared between an updated AI HFpEF model (EchoGo Heart Failure v2) and existing clinical scores (H2FPEF and HFA-PEFF). The AI HFpEF model and H2FPEF score demonstrated similar discrimination and calibration, but classification was higher with AI than H2FPEF and HFA-PEFF, attributable to fewer intermediate scores, due to discordant multivariable inputs. The continuous AI HFpEF model output added information beyond the H2FPEF, and its integration with existing scores increased correct management decisions. Those with a diagnostic positive results from AI had a two-fold increased risk of the composite outcome. We conclude that integrating an AI HFpEF model into the existing clinical diagnostic pathway would improve identification of HFpEF in complex clinical cohorts, and patients at risk of adverse outcomes.

Results – Patient Population (page 9, row 190): “A final sample of 240 cases and 256 were identified after application of exclusions (Figure 1).” The sentence appears to be missing the word “controls”.

Thank you for highlighting this error, it has now been fixed.

Page 9, line 10 now reads:

A final sample of 240 cases and 256 controls were identified after application of exclusions (Figure 1).

Response to Reviewer 2:

Thank you for your ongoing support and motivation to improve the quality of the manuscript. Where appropriate and necessary, we have broken the reviewer’s comments into sub-sections to more completely address the points they highlight.

1. I still feel that findings secondary to categorization/dichotomization are suboptimal or even misleading. Part of my motivation for raising these issues is that I feel the key study findings remain valuable without much of what I perceive to be an over-emphasis on arbitrary categories of continuous scores.

Thank you for this comment. We do tend to agree that the manuscript provides value when only considering the continuous probability estimates, and agree that frequently, there is severe information loss when categorizing continuous values. We feel that changes made throughout the manuscript based on the reviewer's comments highlighting the value of the continuous output of the model support this point further.

However, our motivation to retain the categorization is that this is indeed how the models (both AI and H2FPEF) were developed (as the reviewer has agreed below), and in our understanding and experience of clinical practice, the categorized scores are more commonly used, with continuous measures as supporting evidence.

This is not to say that this should necessarily be the correct way, or “best” way of using the models in practice, but rather an acknowledgement of practice as it actually is and the preference for clinicians to have dichotomous cutoffs for ease of interpretation. As the reviewer highlights below, this approach is also out of necessity, since there are no evidence-based cut-offs indicated in the literature beyond those in the original publication, including the added sensitivity analysis provided based on choice of threshold.

We have strived to ensure that any statistics related to classification (i.e., after a decision threshold has been applied) are not misleading, and have the relevant contextual information to interpret the values (e.g., with or without uncertain data included in the calculation, comparisons at different thresholds, etc). We believe that everything presented is therefore factually correct, and backed up by relevant analysis.

Notwithstanding the value of continuous measurements, as agreed above, this does not invalidate the approach of assessing a model via multiple characteristics; discrimination, classification, calibration, and utility. This is indeed the approach suggested in multiple “guideline” documents (for most comprehensive and recent, please see: <https://www.bmj.com/content/384/bmj-2023-074820>). Given the recent publication of the TRIPOD-AI guidelines (<https://www.bmj.com/content/385/bmj-2023-078378>), also advocating for a holistic view of performance analysis, we have added additional sections and reframed areas of the manuscript to ensure that we are consistent with best practice (see **Supplemental Table 1**).

Supplemental Table 1. TRIPOD-AI and TRIPOD-AI Abstract Checklist

Section/Topic	Item	Development/ Evaluation	Checklist Item	Page
TITLE				
<i>Title</i>	1	Dev+Eval	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted	1
ABSTRACT				
<i>Abstract</i>	2	Dev+Eval	TRIPOD+AI for Abstracts Checklist	2
<i>Abstract: Title</i>	2a	Dev+Eval	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted	2
<i>Abstract: Background</i>	2b	Dev+Eval	Provide a brief explanation of the healthcare context and rationale for developing or evaluating the performance of all models	2
<i>Abstract: Objectives</i>	2c	Dev+Eval	Specify the study objectives, including whether the study describes model development, evaluation, or both	2
	2d	Dev+Eval	Describe the sources of data	2
	2e	Dev+Eval	Describe the eligibility criteria and setting where the data were collected	2
	2f	Dev+Eval	Specify the outcome to be predicted by the model, including time horizon of predictions in case of prognostic models	2
<i>Abstract: Methods</i>	2g	Dev+Eval	Specify the type of model, a summary of the model-building steps, and the method for internal validation	2
	2h	Dev+Eval	Specify the measures used to assess model performance (eg, discrimination, calibration, clinical utility)	2
	2i	Dev+Eval	Report the number of participants and outcome events	2
<i>Abstract: Results</i>	2j	Dev+Eval	Summarise the predictors in the final model	NA
	2k	Dev+Eval	Report model performance estimates (with confidence intervals)	2
<i>Abstract: Discussion</i>	2l	Dev+Eval	Give an overall interpretation of the main results	2
<i>Abstract: Registration</i>	2m	Dev+Eval	Give the registration number and name of the registry or repository	NA

INTRODUCTION				
<i>Background</i>	3a	Dev+Eval	Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing	3-4
	3b	Dev+Eval	or evaluating the prediction model, including references to existing models	
			Describe the target population and the intended purpose of the prediction model in the context of the	3-4
			care pathway, including its intended users (e.g., healthcare professionals, patients, public)	
<i>Objectives</i>	3c	Dev+Eval	Describe any known health inequalities between sociodemographic groups	-
	4	Dev+Eval	Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both)	3-4
METHODS				
<i>Data</i>	5a	Dev+Eval	Describe the sources of data separately for the development and evaluation datasets (e.g., randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data	4-6
	5b	Dev+Eval	Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up	4-5
<i>Participants</i>	6a	Dev+Eval	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including the number and location of centres	4-5
	6b	Dev+Eval	Describe the eligibility criteria for study participants	5
	6c	Dev+Eval	Give details of any treatments received, and how they were handled during model development or evaluation, if relevant	NA
<i>Data Preparation</i>	7	Dev+Eval	Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups	Figure 1 and ST2
<i>Outcome</i>	8a	Dev+Eval	Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups	5
	8b	Dev+Eval	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors	4-5
<i>Predictors</i>	8c	Dev+Eval	Report any actions to blind assessment of the outcome to be predicted	4-5
	9a	Dev	Describe the choice of initial predictors (e.g., literature, previous models, all available predictors) and any pre-selection of predictors before model building	NA
	9b	Dev+Eval	Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors)	6-7
	9c	Dev+Eval	If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors	NA
<i>Sample Size</i>	10	Dev+Eval	Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation	Supplemental Methods
<i>Missing Data</i>	11	Dev+Eval	Describe how missing data were handled. Provide reasons for omitting any data	10, Supplemental Methods, ST1
<i>Analytical Methods</i>	12a	Dev	Describe how the data were used (e.g., for development and evaluation of model performance) in the analysis, including whether the data were partitioned, considering any sample size requirements	NA
	12b	Dev	Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation).	NA
	12c	Dev	Specify the type of model, rationale, all model-building steps, including any hyperparameter tuning, and method for internal validation	NA
	12d	Dev+Eval	Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters (e.g., hospitals, countries). See TRIPOD-Cluster for additional considerations	NA
	12e	Dev+Eval	Specify all measures and plots used (and their rationale) to evaluate model performance (e.g., discrimination, calibration, clinical utility) and, if relevant, to compare multiple models	7-9 10-13 Fig 2-4
	12f	Eval	Describe any model updating (e.g., recalibration) arising from the model evaluation, either overall or for	Supplemental

			particular sociodemographic groups or settings	Metho ds NA
	12g	Eval	For model evaluation, describe how the model predictions were calculated (e.g., formula, code, object, application programming interface)	
<i>Class imbalance</i>	13	Dev+Eval	If class imbalance methods were used, state why and how this was done, and any subsequent methods to recalibrate the model or the model predictions	NA
<i>Fairness</i>	14	Dev+Eval	Describe any approaches that were used to address model fairness and their rationale	-
<i>Model Output</i>	15	Dev	Specify the output of the prediction model (e.g., probabilities, classification). Provide details and rationale for any classification and how the thresholds were identified	NA
<i>Training versus evaluation</i>	16	Dev+Eval	Identify any differences between the development and evaluation data in healthcare setting, eligibility criteria, outcome, and predictors	5-6
<i>Ethical Approval</i>	17	Dev+Eval	Name the institutional research board or ethics committee that approved the study and describe the participant-informed consent or the ethics committee waiver of informed consent	4
OPEN SCIENCE				
<i>Funding</i>	18a	Dev+Eval	Give the source of funding and the role of the funders for the present study	22
<i>Conflicts of Interest</i>	18b	Dev+Eval	Declare any conflicts of interest and financial disclosures for all authors	22
<i>Protocol</i>	18c	Dev+Eval	Indicate where the study protocol can be accessed or state that a protocol was not prepared	4
<i>Registration</i>	18d	Dev+Eval	Provide registration information for the study, including register name and registration number, or state that the study was not registered	4
<i>Data Sharing</i>	18e	Dev+Eval	Provide details of the availability of the study data	18
<i>Code Sharing</i>	18f	Dev+Eval	Provide details of the availability of the analytical code	18
PATIENT & PUBLIC INVOLVEMENT				
<i>Patient and Public Involvement</i>	19	Dev+Eval	Provide details of any patient and public involvement during the design, conduct, reporting, interpretation, or dissemination of the study or state no involvement	22
RESULTS				
<i>Participants</i>	20a	Dev+Eval	Describe the flow of participants through the study, including the number of participants with and without the outcome and, if applicable, a summary of the follow-up time. A diagram may be helpful.	Fig1
	20b	Dev+Eval	Report the characteristics overall and, where applicable, for each data source or setting, including the key dates, key predictors (including demographics), treatments received, sample size, number of outcome events, follow-up time, and amount of missing data. A table may be helpful. Report any differences across key demographic groups.	9 Table1 ST4
	20c	Eval	For model evaluation, show a comparison with the development data of the distribution of important predictors (demographics, predictors, and outcome)	ST5 and ST6
<i>Model development</i>	21	Dev+Eval	Specify the number of participants and outcome events in each analysis (e.g., for model development, hyperparameter tuning, model evaluation)	9
<i>Model Specification</i>	22	Dev	Provide details of the full prediction model (e.g., formula, code, object, application programming interface) to allow predictions in new individuals and to enable third-party evaluation and implementation, including any restrictions to access or re-use (e.g., freely available, proprietary)	NA
<i>Model Performance</i>	23a	Dev+Eval	Report model performance estimates with confidence intervals, including for any key subgroups (e.g., sociodemographic). Consider plots to aid presentation.	10-13 Figs 2-4
	23b	Dev+Eval	If examined, report results of any heterogeneity in model performance across clusters. See TRIPOD	NA
<i>Model Updating</i>	24	Eval	Cluster for additional details Report the results from any model updating, including the updated model and subsequent performance	Supp Metho ds
DISCUSSION				
<i>Interpretation</i>	25	Dev+Eval	Give an overall interpretation of the main results, including issues of fairness in the context of the objectives and previous studies	14-19
<i>Limitations</i>	26	Dev+Eval	Discuss any limitations of the study (such as a non-representative sample, sample size, overfitting, missing data) and their effects on any biases, statistical uncertainty, and generalizability	18-19
<i>Useability of the model in the context of current care</i>	27a	Dev	Describe how poor quality or unavailable input data (e.g., predictor values) should be assessed and handled when implementing the prediction model	NA
	27b	Dev	Specify whether users will be required to interact in the handling of the input data or use of the model, and what level of expertise is required of users	NA
	27c	Dev+Eval	Discuss any next steps for future research, with a specific view to applicability and generalizability of the model	14-19

LEGEND: Shown is the completed TRIPOD-AI and TRIPOD-AI Abstract Checklist from Collins et al., TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *British Medical Journal*. 2024; 385. Dev = items relevant only to the development of a prediction model; Eval = items relating solely to the evaluation of a prediction model; Dev+Eval = items applicable to both the development and evaluation of a prediction model **Source:** <https://www.bmj.com/content/bmj/suppl/2024/04/17/bmj-2023-078378.DC5/colg078378.wt2.pdf>
AI+Tripod For Abstracts Source: <https://www.tripod-statement.org/wp-content/uploads/2019/12/TRIPODAI-for-Abstracts.pdf>

Moreover, we have modified the limitations in the text to reflect the role that the continuous probability outputs and raw score values may have in providing additional granularity in risk.

Page 19, line 12 now reads:

Third, the continuous outcome probability of HFpEF from the AI HFpEF model, similar to the raw score values for the H2FPEF score, may provide additional information on risk beyond the dichotomized results.

The H2FPEF score is a continuous score with a nearly identical AUROC to AI HFpEF. Therefore, claims of “higher classification performance” are not sufficiently justified.

Thank you for this comment. We acknowledge that the AUROC for AI HFpEF model and H2FPEF are indeed very similar, however we would respectfully disagree that “higher classification performance” is not justified.

Firstly, as we are sure the reviewer is aware, there is a difference between discrimination (AUROC) and classification (Se/Sp/NPV/PPV), with each performance metric highlighting different characteristics of the models. Thus, notwithstanding a similar AUROC, classification can still differ between models.

Furthermore, this inferential statement is supported by relevant and appropriate statistical tests (i.e., proportions test). We have now highlighted this in the main text.

Page 8, line 4 now reads:

*Classification performance, including the sensitivity, specificity, negative predictive value (NPV), and positive predictive value (PPV) was determined for all models (**Supplemental Table 3**). Proportions tests are used for statistical comparisons between models for classification performance.*

As highlighted in the previous round of reviews, we have strived to ensure that the classification statistics are all appropriately presented, such that a reader can interpret whether their calculation includes uncertain data (or not). Furthermore, the presentation of the method for calculation (*see answer to 1b below*), and all relevant confusion matrix values (**Supplemental Table 7**), ensure that an interested reader can replicate these values and associated statistical tests.

Furthermore, we believe we have contextualized the nuance in extensive discussion on this topic below (**bold highlighted for emphasis**):

Page 16, line 1 now reads:

In the current study, applying an updated version model to an independent validation cohort, the AI HFpEF model had similar performance to multiparametric clinical scores to identify HFpEF, with an AUROC of 0.798 (95% CI: 0.756-0.799) compared to 0.788 (0.745-0.798) for the H2FPEF score, but resulted in fewer

intermediate classifications, with only 15.1% having a non-diagnostic classifications compared to 61.7% and 54.2% of those classified according to the H2FPEF and HFA-PEFF scores, respectively. **Consequently, classification performance was higher using the AI HFpEF model than clinical scores when considering these intermediate classifications, but lower when patients with intermediate classifications were removed.** This supports previous investigations^{12,13} demonstrating that existing clinical scores have excellent performance (particularly sensitivity) when they provide a clear diagnostic output, but are limited by high intermediate classification rates, which might be more apparent in situations of low data quality (e.g., missing, poor imaging quality) or true clinical uncertainty (discordant parameters). Sub-groups analysis of patients in whom the multiparametric models were all available (i.e., retaining only clinical discordance), demonstrated improved clinical score performance, but retained high rates of intermediate classifications (61% for H2FPEF, 52% for HFA-PEFF), indicating that clinical discordance remains a critical issue during the diagnostic pathway.

Higher sensitivity/specificity using AI-HFpEF versus H2FPEF is based purely on the operating points evaluated, and would differ across other operating points on the ROC curve (and on average, we know the ROC curves are the same, so there are also clearly points where sens/spec are higher for H2FPEF). Although I don't think it makes a very strong argument, I think the authors can elect to keep the current evaluation at the cutpoints provided because the cutoffs are indeed mentioned in the H2FPEF paper, but the comparisons need to be placed in a more explicit and nuanced context that the comparison model could easily be compared at other cutpoints (and indeed, when they are compared using deciles of probability instead, performance is nearly identical, as expected). It would perhaps be another story if there were firm medical consensus or guideline statements promoting a H2FPEF score of 6-9 to make treatment decisions, but I just don't see that as the case. For HFA-PEFF, which is not a continuous score, the comparison made is less problematic, but the analysis would still benefit from more context on the cutpoints themselves.

We agree with the reviewer that the cutpoints chosen are the most appropriate with the current evidence and original development of the clinical model. While we generally disagree that “performance is nearly identical [at other cutpoints]” due to the holistic and nuanced nature of “performance”, we understand that classification performance can be assessed as a multitude of decision thresholds. We hope that the contextual information provided throughout is sufficient to indicate that this might not be the optimal approach in practice, and further testing is required to elucidate the “optimal” approach. Indeed, we would like to highlight to the reviewer that our main conclusions and recommendations for future research and clinical use would be to combine the information available, rather than relying on a simple like-for-like comparison and suggestions of one “best” diagnostic tool.

Page 17, line 6 now reads:

Suggestions to integrate AI models into existing diagnostic pathways^{16,17} and utilize the combined clinical information from all available sources are supported by the current study. In decision modelling, use of combined information from the AI HFpEF model and clinical scores resulted in a greater number of individuals appropriately receiving treatment (in this case SGLT2 inhibitors) with fewer number of individuals treated unnecessarily. This is logical, given that each model is associated with its own development process (AI, statistical modelling, and clinical consensus for AI HFpEF, H2FPEF, and HFA-PEFF, respectively). In a heterogeneous syndrome like HFpEF, each model would likely capture different types of patients under the broad umbrella of HFpEF. This feature becomes especially important given the absence of one single standard definition of HFpEF, and heterogeneity of the etiology of the syndrome; different diagnostic models which can be readily integrated would capitalize on such ambiguity and provide the greatest value for patients. The continuous output of the AI HFpEF model added significant value to the information contained in the existing clinical scores, which is likely due to the improved AI HFpEF model calibration (beyond that apparent in version 1). However, whether the continuous or categorical output would be used in clinical practice has yet to be prospectively evaluated, and exactly how different models and

associated decision thresholds may be combined requires extensive validation. This is important, particularly considering that the greatest benefit was observed when all models were utilized for identification of disease and decision-making processes.

In practice, patients with undifferentiated dyspnea could be screened using a clinical score or AI model and, if inconclusive/intermediate, alternative models could be applied to identify those that are truly at lower risk versus those with more uncertainty for which confirmatory testing (e.g. stress testing or invasive testing with right heart catheterization) could be performed. While this strategy requires prospective testing to evaluate its impact on care patterns and outcomes, results from this validation study suggest promise of a combined clinical and AI approach to improve upon existing HFpEF risk stratification, and hopefully reduce the burden of under-recognition in clinical practice.³⁹ A possible strategy for future deployment and implementation might require an evaluation of the optimal decision thresholding strategy across multiple clinical models, such that each model capitalizes on its strengths, and supports any weaknesses of others.

b. On a related note, I find the criteria used to define the categories of AI-HFpEF not sufficiently justified. How were these very particular (e.g., uncertainly less than 0.44, or prediction class probability greater than or equal to 0.5) cutpoints derived? This may be explained in the prior papers, but some more context in this paper/supplement is needed.

We apologize, this was indeed detailed in other related manuscript, but as suggested we have now added the following text to the supplement.

Supplemental Methods now read:

The choice of decision thresholds for prediction class probability (0.5), instability (0.20), and uncertainty (0.44), are based on bench testing during algorithmic development (training and validation). For prediction class probability, 0.5 was chosen as a default threshold wherein the model would be numerically closer to predicting case than control (or vice versa; i.e., the middle value in the probability continuum). Multiple measurements of uncertainty and instability were tested to assess which parameter was most commonly associated with incorrect predictions. Experiments were completed wherein 5 to 45% of the most uncertain data were removed, and model accuracy, sensitivity, and specificity were assessed. The curves for performance metrics as a function of data removal monotonically increase with more data removed. Given a maximum tolerable reduction in data of 10% (i.e., proportion of data removed due to uncertainty and/or instability), thresholds for these metrics were identified wherein there were diminishing returns in performance for increasing data removal, without exceeding the 10% threshold. Finally, multiple combinations of thresholds between the two metrics were compared to ensure the highest possible performance without exceeding the 10% threshold of data removal.

2. How are the test characteristics (e.g., sensitivity, specificity) calculated with respect to the ‘non-diagnostic’ categories? Is ‘non-diagnostic’ considered a negative test result? A robust way to handle this would be to calculate test characteristics considering all ‘non-diagnostic’ results are negative, and conversely all of them as positive.

We apologize for this omission; we have added a detailed breakdown of the calculation to the supplement.

Supplemental Methods, section on Calculation of Classification Statistics, now reads:

All three clinical scores provided a “non-diagnostic” classification when there was discordant, or missing information to reliably suggest HFpEF presence or absence. Classification performance is therefore presented when considering all patients (i.e., including non-diagnostic results in the calculation of classification statistics), and when considered only diagnostic outputs (i.e., positive or negative). The calculation of these statistics are described below.

Supplemental Table 3. Classification Statistics Methodology

<i>Model Output</i>	<i>Ground Truth Positive</i>	<i>Ground Truth Negative</i>	<i>Total</i>	<i>Predictive Performance (%)</i>
<i>Positive</i>	<i>A</i>	<i>D</i>	<i>T1</i>	PPV (%) : $(A/T1)*100$
<i>Negative</i>	<i>B</i>	<i>E</i>	<i>T2</i>	NPV (%) : $(B/T2)*100$
<i>No Classification</i>	<i>C</i>	<i>F</i>	<i>T3</i>	
<i>Total</i>	<i>N1</i>	<i>N2</i>	<i>N</i>	
<i>Diagnostic Performance (Intended Use)</i>	Sensitivity (%) $(A/(A+B))*100$	Specificity (%) $(E/(D+E))*100$		
<i>Performance (Including No Classification)</i>	Sensitivity (%) $(A/N1)*100$	Specificity (%) $(E/N2)*100$		

LEGEND: Shown is the calculation methodology for classification statistics when removing and including no classification studies in the calculation. PPV, positive predictive value; NPV, negative predictive value.

We have also added additional analyses to highlight the impact of the non-diagnostic results. We have not elected to add all non-diagnostic results as positive, or negative, since we did not believe that this is intuitive or an accurate representation of the non-diagnostic approach. Instead, we have demonstrated performance when only considering the prediction class probability, and not the uncertainty measurement.

Table 2 now reads:

<i>Model</i>	<i>Calculation Method</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>NPV</i>	<i>PPV</i>
<i>AI HFpEF</i>	<i>All Data</i>	77.4% (74.6-79.6%)	50.2% (48.6-52.1%)	81.6% (73.6%-87.6%)	67.3% (59.6%-71.7%)
	<i>Diagnostic Only</i>	86.6% (80.2-90.9%)	58.5% (51.0-64.8%)	81.6% (73.6%-87.6%)	67.3% (59.6%-71.7%)
	<i>Ignore Uncertainty</i>	83.8% (78.2-88.1%)	53.9% (47.5-60.0%)	78.0% (70.3-82.8%)	63.0% (56.1-67.0%)
<i>H2FPEF</i>	<i>All Data</i>	53.9% (50.2-58.2%)	12.8% (11.8-13.9%)	90.3% (75.0-100.0%)	73.6% (64.9-79.3%)
	<i>Diagnostic Only</i>	97.5% (94.0-100.0%)	40.0% (26.8-52.9%)	90.3% (75.0-100.0%)	73.6% (64.9-79.3%)
<i>HFA-PEFF</i>	<i>All Data</i>	63.2% (58.3-68.0%)	29.0% (26.9-31.0%)	98.5% (94.7-100.0%)	86.3% (79.3-91.1%)
	<i>Diagnostic Only</i>	99.3% (97.3-100.0%)	74.7% (62.8-83.8%)	98.5% (94.7-100.0%)	86.3% (79.3-91.1%)

LEGEND: Classification statistics and 95% confidence intervals (CIs) are provided for the three models using all available data (i.e., including those with intermediate classifications in the calculation; See Supplemental Table 3), and when only considering diagnostic studies (i.e., excluding those with intermediate classifications in the calculation). For AI HFpEF model, classification performance is also presented when uncertainty metrics are disregarded, and classification of patients is based only on prediction class probability (i.e., ≥ 0.50). For further information related to the calculation of these statistics, please see **Supplemental Table 3**. AI HFpEF = artificial intelligence model for the prediction of heart failure with preserved ejection fraction, HF2PEF = Heavy, Hypertensive, Atrial Fibrillation, Pulmonary Hypertension, Elderly, Filling Pressure, HFA-PEFF = Heart Failure Association Pre-test assessment, Echocardiography and Natriuretic Peptide Score, Functional Testing in Case of Uncertainty, Final Etiology, NPV = negative predictive value, PPV = positive predictive value

3. I struggle in general with the term ‘non-diagnostic’ here. As the authors mention, there is no real gold standard for HFpEF so none of the scores or their values are ‘diagnostic’ in the first place. Furthermore, the intermediate scores still provide some information. Looking back at the original H2FPEF and HFA-PEFF papers, the term ‘intermediate’ appears to be employed and I think this is a far more appropriate term. Along the same lines, I think there needs to be a clearer distinction between ‘score not calculable’ and ‘intermediate value’, where the former is more truly in a sense ‘non-diagnostic’.

We agree that the word “non-diagnostic” might be better referred to as “intermediate”. We have changed throughout the manuscript and discuss the terminology as follows.

Page 6, line 18 now reads:

A third categorical output, “no classification”, herein referred to as “intermediate,” is generated based on model instability and uncertainty, to ensure that unstable and uncertain predictions are not provided to a user (Supplemental Methods). The term intermediate is used to highlight that important diagnostic information can still be contained within uncertain predictions. The same terminology is also used for the multiparametric clinical scores described below.

We feel that the distinction between “discordance” and “missingness” is indeed already highlighted throughout the manuscript and in the supplement. However, “score not calculable” – in the reviewer’s terms – is more complex, since neither of the clinical score provide guidance on when the score can or cannot be used given the missingness of clinical parameters.

If one assumes that a score is calculable when there are any of the parameters available, then this is the case for all patients (e.g., **Supplemental Figure 1**). Indeed, it is not possible to define how many parameters must be available – or how many must be missing – for this label to hold validity; any assumption on this matter is likely conjecture, and certainly not in keeping with the publications from which the clinical model is based.

Instead, we have demonstrated performance characteristics across the range of scores, and for those with full parameters compared with missing parameter inputs. These additions in the previous round of revisions highlight that “intermediate” can be achieved by either missingness, discordance, or a combination of the two. The most conservative approach, and favorable to the existing clinical scores, is to only calculate performance in those with all parameters available, which would be considered the “ideal” approach.

4. On a related note, in the discussion the authors seem to equate intermediate score values with ‘conflicting’ clinical information, but that is not necessarily the same. Someone may have an intermediate H2FPEF score simply because they are young...I don’t think that is necessarily in ‘conflict’ with the other risk factors.

Thank you for this comment. We understand the confusion here, but assuming that all other parameters are not indicative of HFpEF, a young patient would not necessarily result in an intermediate score. Age only contributes 1 point to the overall score, and thus if all other parameters were “positive” (i.e., indicating HFpEF), and a patient was young, their score would be 8, and thus still indicative of HFpEF.

Instead, assuming all parameters are available, a more common and likely scenario is that multiple scores are discordant, resulting in intermediate probabilities. The nuance between intermediate scores and the source of this score is documented in the main manuscript and supplement; an intermediate score can be due to discordance or missingness. The impact of these two possibilities is presented as highlighted above (Comment on reviewer point #3).

5. More information is needed regarding the NRI analyses. I am very surprised to see highly positive NRI for AI-HFpEF versus H2PEF (NRI 0.40 [0.23-0.57] and IDI 0.07 [0.01-0.13]) when the AUROCs are the same. Further, the NRI shown in Fig 3 (0.03 [-0.06-0.20]) is very difficult and more expected. I'm unclear on what is different. Also, it is necessary to provide both the NRI+ and NRI- components given they imply very different things.

Firstly, we apologize for the omission of the events (+) and nonevent (-) NRI, it was originally intended to be included in the supplement but was mistakenly omitted. It has now been added. Please see **Supplemental Table 17**, reproduced for convenience below.

Supplemental Table 17. *Reclassification Statistics for Continuous and Categorical Model Outputs*

Comparison	NRI	NRI (+)	NRI (-)	IDI	LRT
Continuous					
<i>AI HFpEF vs. H2FPEF</i>	0.402 (0.209-0.589)	0.183 (0.042-0.313)	0.219 (0.093-0.346)	0.070 (0.003-0.137)	<i>P</i> <0.001
Categorical					
<i>AI HFpEF vs. H2FPEF</i>	0.018 (-0.063-0.194)	-	-	-	<i>P</i> <0.001
<i>AI HFpEF vs. HFA-PEFF</i>	0.167 (0.005-0.330)	-	-	-	<i>P</i> <0.001

LEGEND: Shown are the added value statistics comparing the AI HFpEF model to the H2FPEF and HFA-PEFF scores. NRI, net reclassification improvement; NRI (+), NRI events; NRI (-) NRI non-events; IDI, integrated discrimination improvement; LRT, Likelihood Ratio Test. Bootstrap 95% confidence intervals are provided.

Hopefully the table presented conveys two key points; (i) presentation of NRI is with respect to comparisons between continuous or categorical model outputs, and (ii) there was a rounding and transcription error with respect to some of the NRI values presented. We feel that the distinction between continuous and categorical comparisons is clear from the original results text, however we have adjusted the values in text and **Figure 3** for correctness, and added clarification where possible.

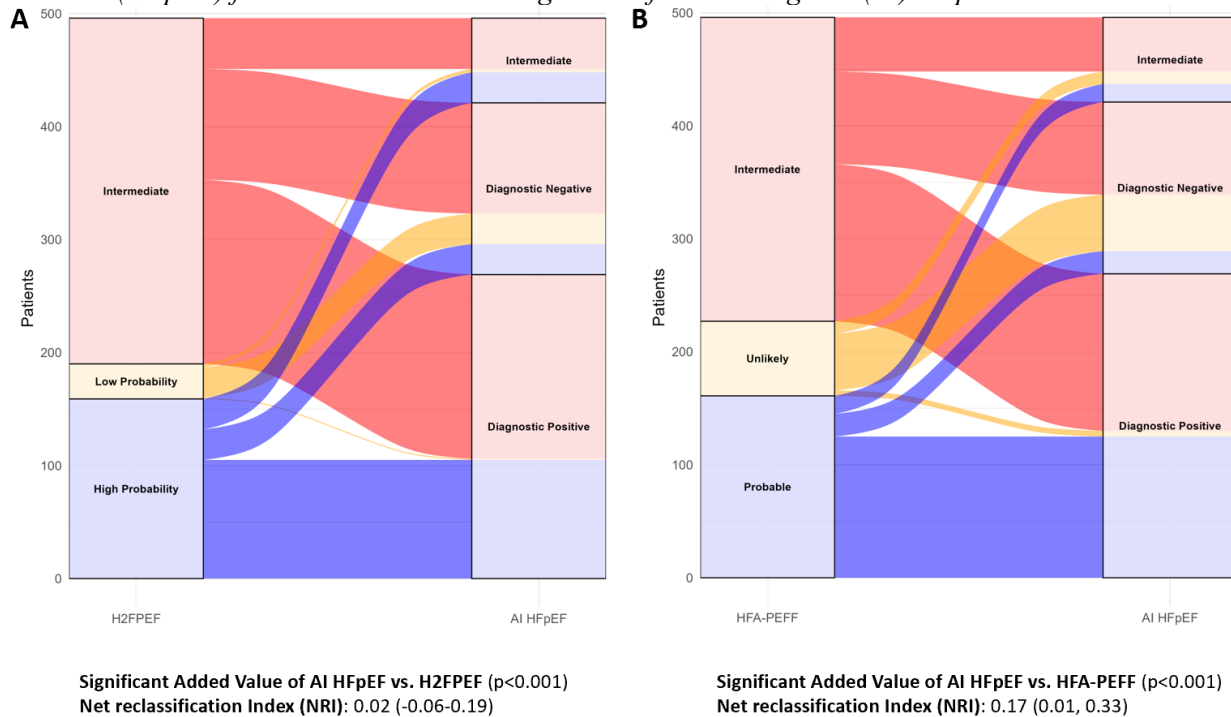
Page 12, line 22 now reads:

Figure 3 illustrates the reclassification of patients from the H2FPEF Score (panel A) and HFA-PEFF Score (panel B) into the AI HFpEF model, alongside net classification statistics for the AI HFpEF model compared to each score. In the intermediate classified patients according to the H2FPEF score (n=306), the AI HFpEF model also classified 45 as intermediate, 163 as high likelihood of HFpEF, and 98 as low likelihood of HFpEF. In the intermediate classified patients according to the HFA-PEFF score (n=269), the AI HFpEF model also classified 48 as intermediate, 139 as high likelihood of HFpEF, and 82 as low likelihood of HFpEF. For the continuous probability output from the AI HFpEF model and the H2FPEF score, the AI HFpEF model provided significantly more information (likelihood ratio test, *p*<0.001), supported by positive overall net reclassification statistics (NRI 0.40, 0.21-0.59; **Supplemental Table 17**), and marginally positive integrated discrimination improvement (IDI) index (0.07, 0.00-0.14). However, when utilizing only the categorical outputs from the three models, the AI HFpEF model added significant information to both the H2FPEF and HFA-PEFF scores according to likelihood ratio tests (both *p*<0.001), but NRI statistics were less consistent (**Figure 3**).

Figure 3 is presented below. We have also added additional text to the legend to ensure clarity as highlighted by the reviewer. For transparency regarding the transcription errors, the main changes are also highlighted here:

- In Figure 3, panel A, NRI was reported as 0.03 (-0.06, 0.20). This has been changed to 0.02 (-0.06, 0.19).
- In Figure 3, panel B, NRI was reported as 0.26 (-0.04, 0.33). This has been changed to 0.17 (0.01, 0.33)

Figure 3. Alluvial Plots Demonstrating Reclassification of Predicted Heart Failure with Preserved Ejection Fraction (HFpEF) from Clinical Scores Using the Artificial Intelligence (AI) HFpEF Model



LEGEND: Displayed are alluvial plots depicting the reclassification of predicted HFpEF from clinical scores using the AI HFpEF model. This plot and associated reclassification statistics account for only categorical classification outputs from each model, rather than continuous outputs. Panel A depicts the AI model's reclassification of an individual's predicted HFpEF status from the H2FPEF score. Panel B depicts the AI model's reclassification of an individual's predicted HFpEF status from the HFA-PEFF score. The added value of the AI HFpEF model compared with the H2FPEF score and HFA-PEFF score are displayed below the alluvial plots. Likelihood ratio test p -values are presented alongside net reclassification improvement (NRI) statistics. All NRI statistics are based on categorical outputs. Non-diagnostic and indeterminate outputs are referred to as "intermediate" for consistency and clarity.

With respect to the interpretation of the NRI and IDI values, we feel that the IDI values are indeed reflective of the AUROC data, since this measure is more reflective of discrimination (*see references below*). With respect to the interpretation of the NRI data, we believe that the changes made in the results section (see above), and abstract and discussion (see below) address the concerns of the reviewer, and are more closely aligned with their views on the continuous vs. categorical use of the models.

Abstract, Page 2, line 12 now reads:

The continuous AI HFpEF model output added information beyond the H2FPEF, and its integration with existing scores increased correct management decisions.

Page 14, line 17 now reads:

In this retrospective case-control study, an AI HFpEF model using deep learning of a single TTE video clip had higher classification performance than existing clinical scores, largely owing to fewer intermediate classifications. When clinical scores provided a diagnostic output, they demonstrated excellent classification performance. The continuous output of the AI HFpEF model demonstrated a significant increase in the information available for diagnosis, beyond that of the clinical scores, but this was less clear for categorical outputs. Overall, combining use of AI and clinical score information resulted in greater rates of appropriate treatment for HFpEF and fewer unnecessary treatments in decision modelling. All algorithms, including both the AI HFpEF model and clinical scores, demonstrated significant associations with risk of all-cause mortality and HF hospitalization. Overall, these results indicate a possible role for this AI HFpEF model in the HFpEF diagnostic pathway, particularly in combination with information from clinical scores, to identify patients warranting confirmatory testing or management for HFpEF

Page 14, line 20 now reads:

The continuous output of the AI HFpEF model demonstrated a significant increase in the information available for diagnosis, beyond that of the clinical scores, but this was less clear for categorical outputs.

Selected Publications on the shortcomings of reclassification statistics and suggested gold-standard approach (LRT)

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC5837334/>
- <https://pubs.rsna.org/doi/full/10.1148/radiol.222343>
- <https://diagnprognres.biomedcentral.com/articles/10.1186/s41512-018-0037-2>
- <https://onlinelibrary.wiley.com/doi/abs/10.1002/sim.5804>

6. Since both scores are mis-calibrated, the overall NRI can be misleading. A percentile-based NRI would be a valuable addition (<https://pubmed.ncbi.nlm.nih.gov/29304237/>) to assess whether the NRI may be driven by asymmetric mis-calibration.

For the reasons highlighted above, and in the referenced literature, we are hesitant to add more reclassification statistics beyond those currently presented. The NRI and IDI can indeed be misleading, and are supported by the gold-standard approach (LRT). However, more challenging with the suggested approach, is the selection of specific percentiles of risk, since these are not suggested in any previous literature on the H2FPEF score. Thus, to avoid comparisons which are potentially misleading, and at worst, invalid in the context of the originally publications (as agreed by the reviewer; see Reviewer Comment #1), we are hesitant to add a percentile-based NRI.

Furthermore, we believe that an exaggerated focus on net reclassification improvement between clinical models is actually contrary to the main point of the manuscript; integration of multiple models (AI and clinical) is potentially important, with each contributing when others have shortcomings (e.g., decision curve analysis). We believe that this approach is not only the fairest and most accurate, but also consistent with a holistic approach to model assessment as per current guidelines (please see: <https://www.bmj.com/content/384/bmj-2023-074820>, <https://www.bmj.com/content/385/bmj-2023-078378>).

7. It would be an interesting and potentially fairly valuable secondary analysis to assess whether AI-HFpEF maintains a degree of classification performance among individuals with low clinical score values. This would suggest that the algorithm is picking up things that are not obvious and speaks more directly to added value.

We agree that this analysis would support the existing decision curve analysis with respect to where the AI model might demonstrate value. This has now been added to the supplement (**Supplemental Tables 8-13**;

not shown here for brevity) alongside more complete subgroup analyses, which we believe is also valuable in identifying where and when each diagnostic tool, and the combination, might be most useful.

Page 11, line 21 now reads:

Supplemental Table 15 illustrates how performance of the H2FPEF changes as a consequence of the decision threshold. As expected, higher decision thresholds result in lower sensitivity, and higher specificity, and the converse is true for lower decision thresholds. Sensitivity ranged from 3.8% to 98.8%, and specificity ranged from 10.9% to 100%, for thresholds of categorical scores (1-9). **Supplemental Table 9** highlights how the AI HFpEF model might compensate for the drop in performance above a score of 6, and below a score of 4 (sensitivity and specificity respectively).

Page 17, line 12 now reads:

In a heterogeneous syndrome like HFpEF, each model would likely capture different types of patients under the broad umbrella of HFpEF. This feature becomes especially important given the absence of one single standard definition of HFpEF, and heterogeneity of the etiology of the syndrome; different diagnostic models which can be readily integrated would capitalize on such ambiguity and provide the greatest value for patients. The continuous output of the AI HFpEF model added significant value to the information contained in the existing clinical scores, which is likely due to the improved AI HFpEF model calibration (beyond that apparent in version 1). However, whether the continuous or categorical output would be used in clinical practice has yet to be prospectively evaluated, and exactly how different models and associated decision thresholds may be combined, and in what type of patient(s), requires extensive validation. This is important, particularly considering that the greatest benefit was observed when all models were utilized for identification of disease and decision-making processes.

The authors would like to thank the reviewers for these important comments which we feel have strengthened the manuscript. We now hope that the manuscript is suitable for publication in *Nature Communications*.